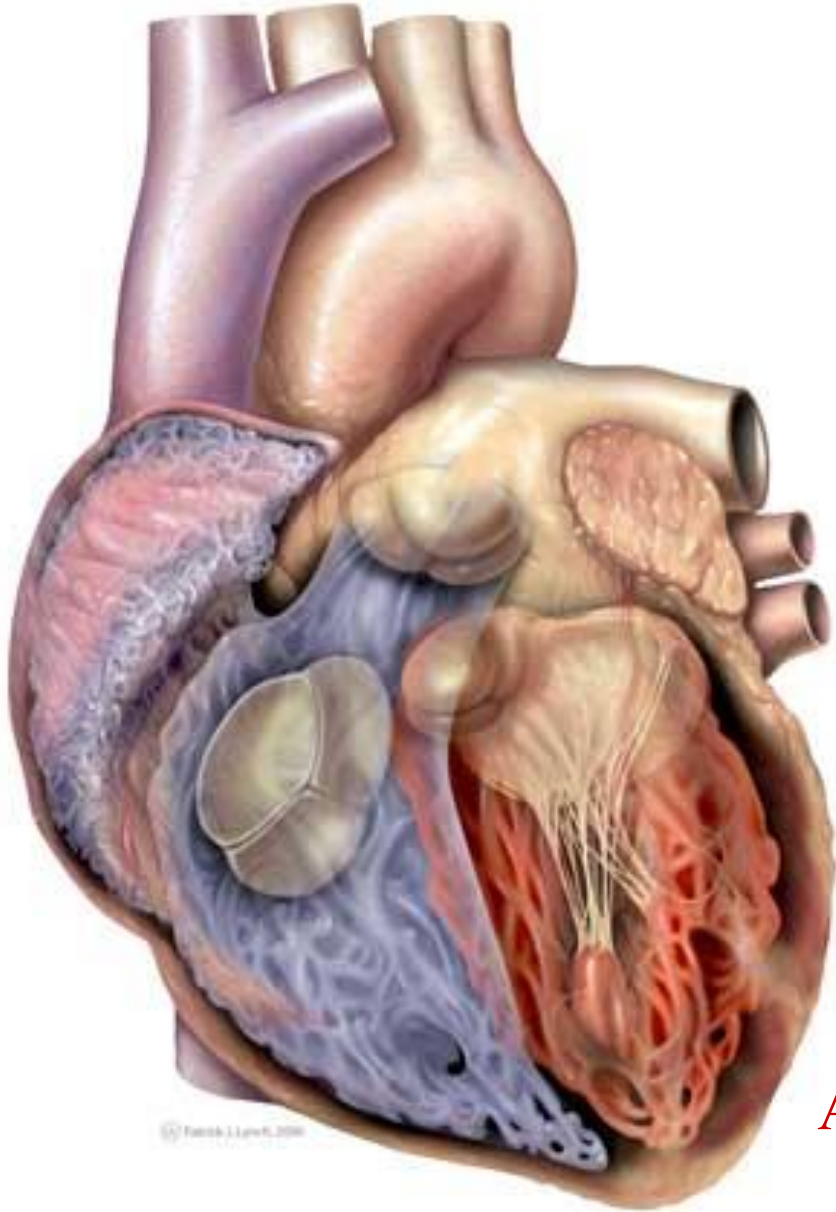




Mediastinum

Anatomy of heart and great vessels

Assoc.Prof.Dr.Pornsawan Duangsuwan

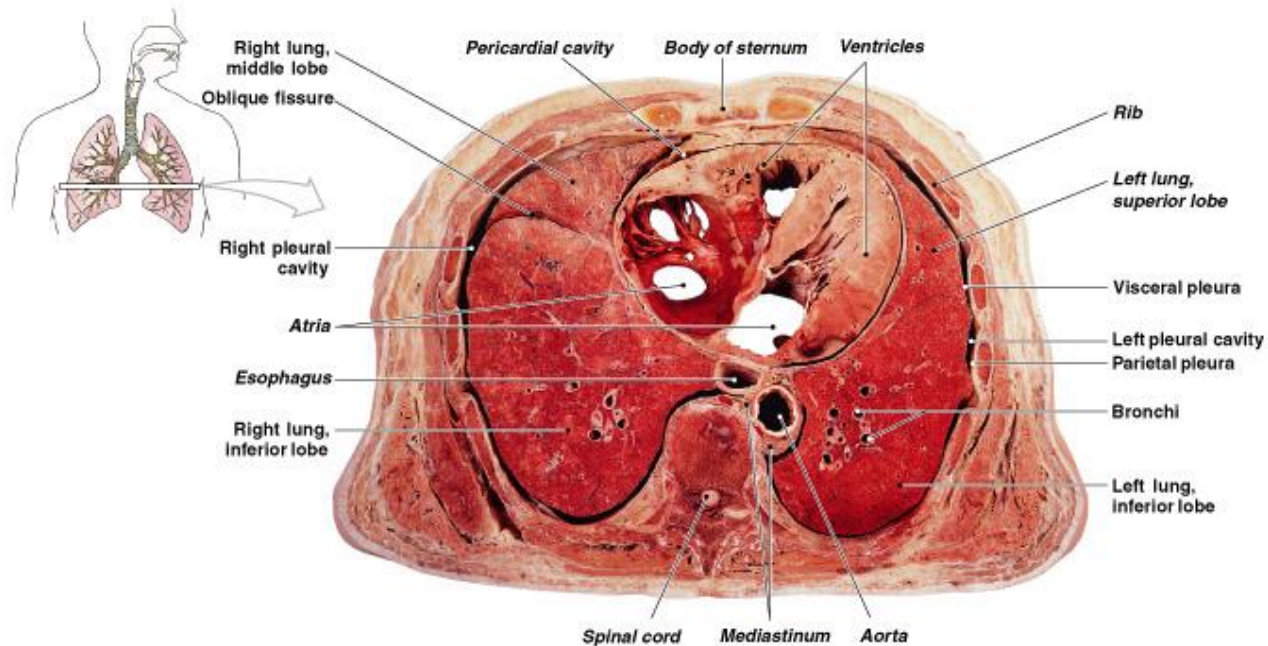
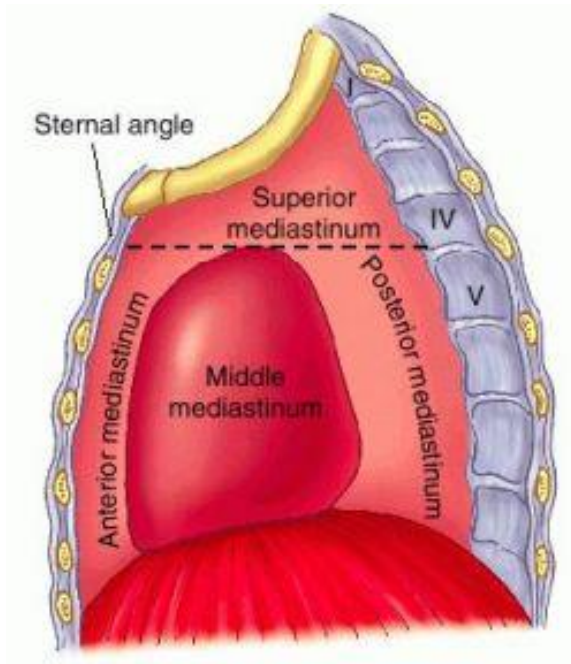
Thoracic cavity

- Left and right pleural cavities
- Mediastinum

Superior mediastinum

Inferior mediastinum

- Anterior mediastinum
- Middle mediastinum
- Posterior mediastinum



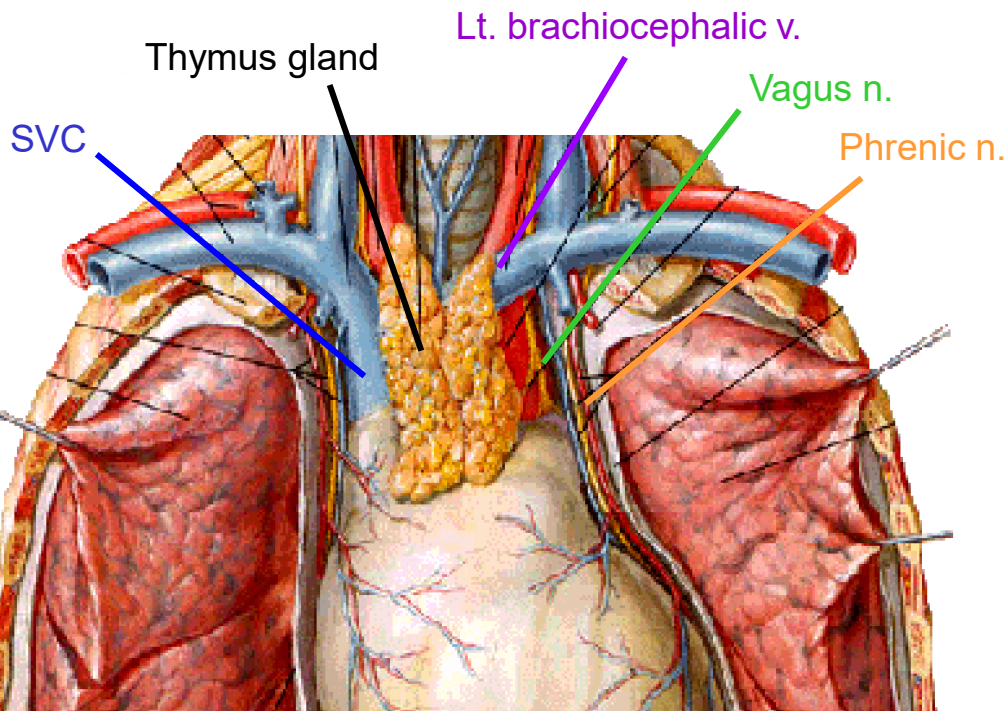
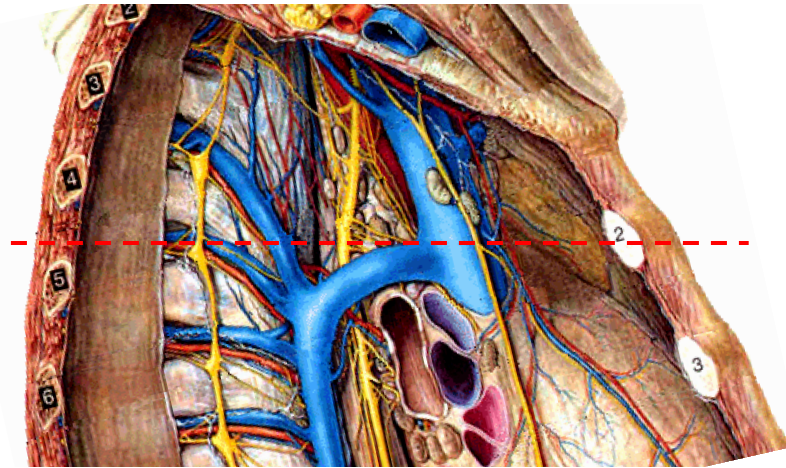
Superior Mediastinum

Boundary

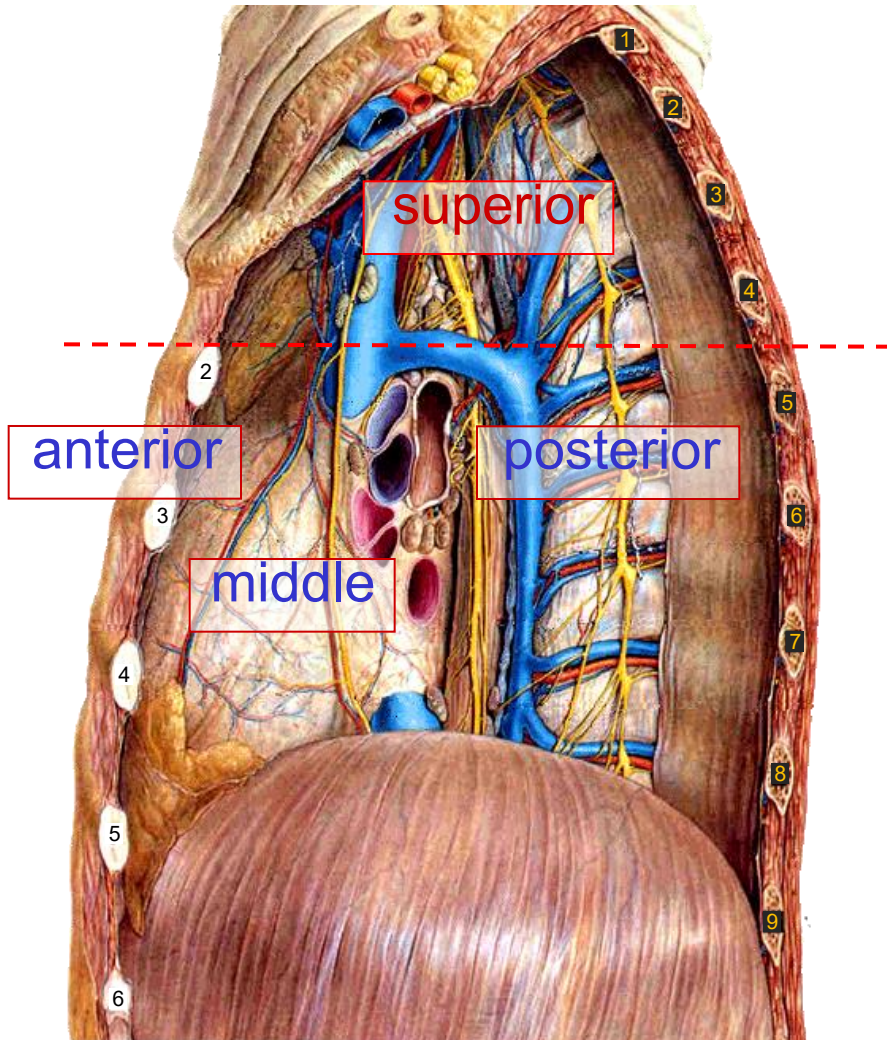
- Superior: thoracic inlet
- Inferior: sternal angle – T4 / T5
- Anterior: manubrium sterni
- Posterior: Thoracic vertebrae 1-4

Contents:

- Thymus gland
- SVC, brachiocephalic v.
- Aortic arch & its 3 branches
- Vagus n., phrenic n.
- Recurrent laryngeal n.
- Esophagus, trachea
- Thoracic duct



Anterior Mediastinum



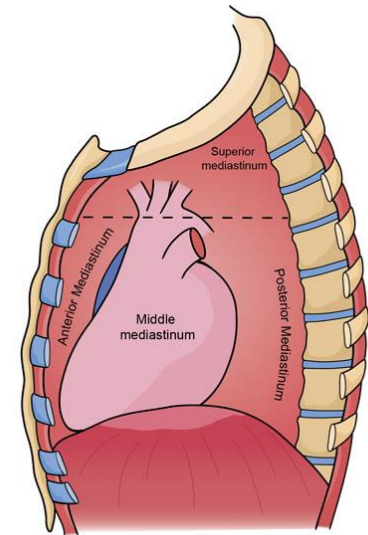
Boundary

- Superior: sternal angle – T4 / T5
- Inferior: diaphragm
- Anterior: body of sternum
- Posterior: pericardium

Contents:

- Fat
- Remnants of thymus gland
- Lymph nodes / vessels
- Sternopericardial ligament
(pericardium – xiphoid process)

Middle mediastinum



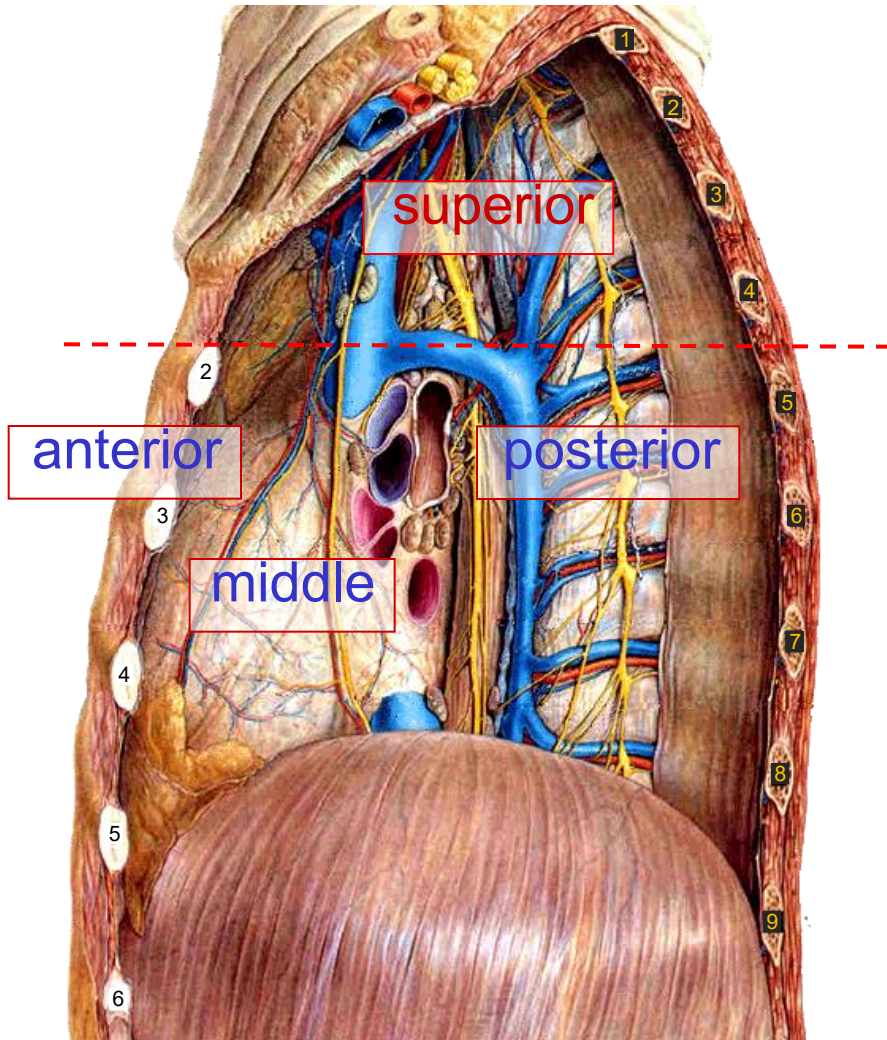
Location

- Between anterior and posterior mediastinum

Contents in middle mediastinum

- Pericardium (pericardial sac)
- Heart and Great vessels
- Phrenic nerves
- Main bronchus

Posterior Mediastinum



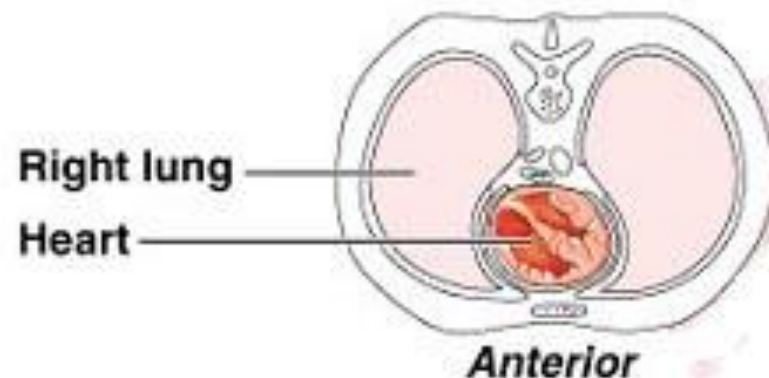
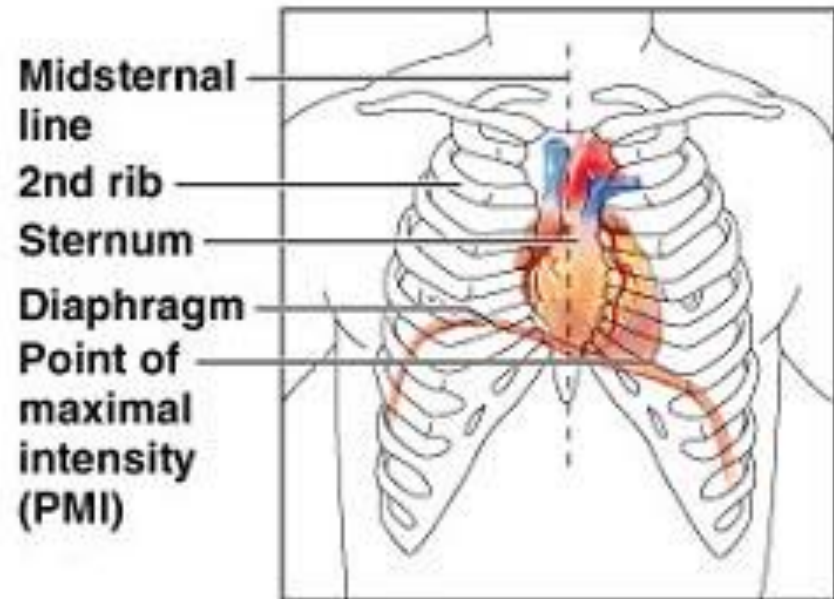
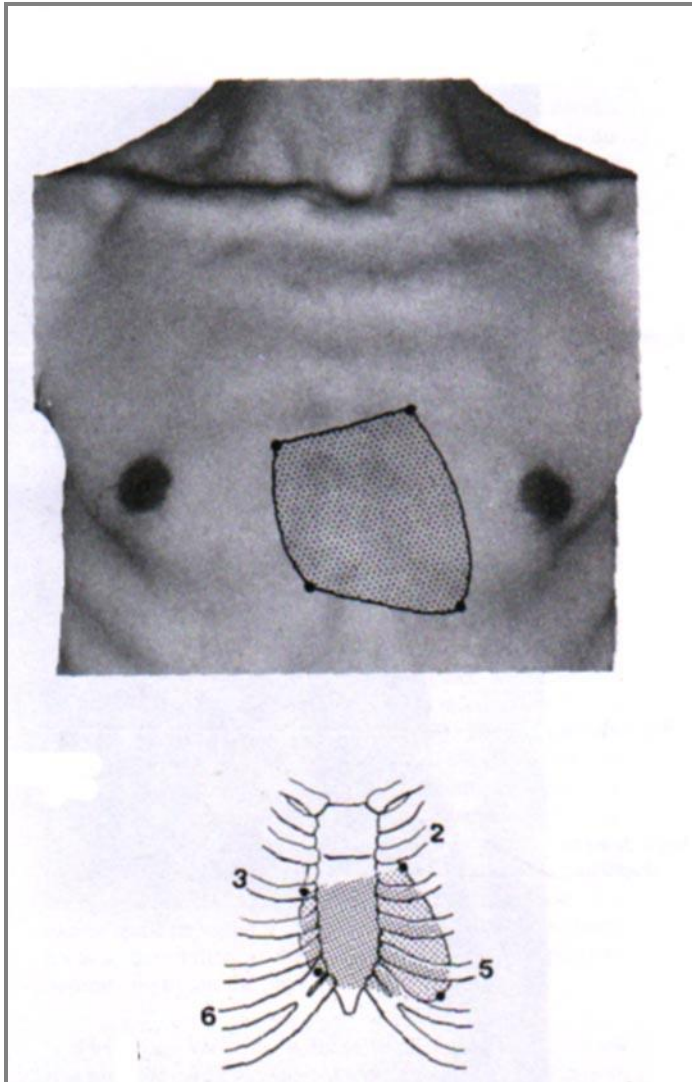
Boundary

- Superior: sternal angle – T4 / T5
- Inferior: diaphragm
- Anterior: pericardium
- Posterior: T5 – T12 vertebrae

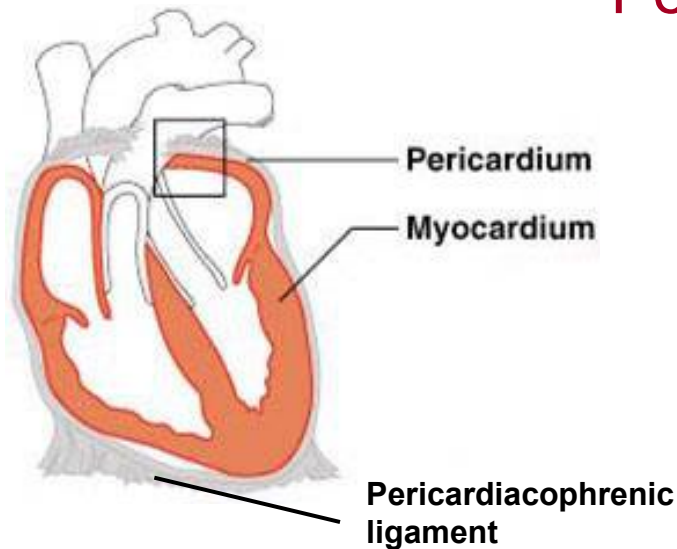
Contents:

- Thoracic aorta
- Thoracic duct
- Posterior mediastinal lymph nodes
- Azygos and hemiazygos veins
- Esophagus, esophageal plexus
- Thoracic sympathetic trunks
- Thoracic splanchnic nerves

SDL: Location of Heart in Thorax

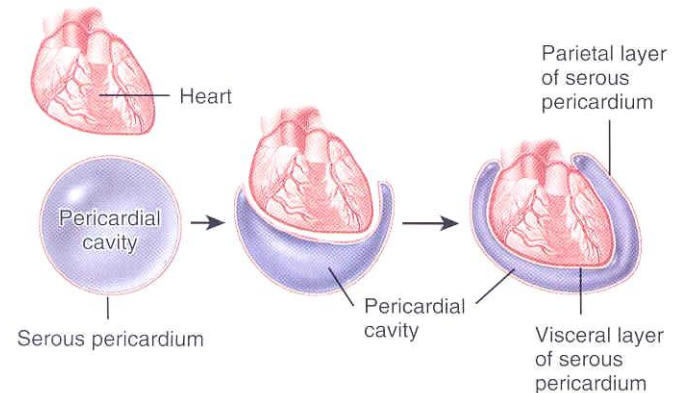
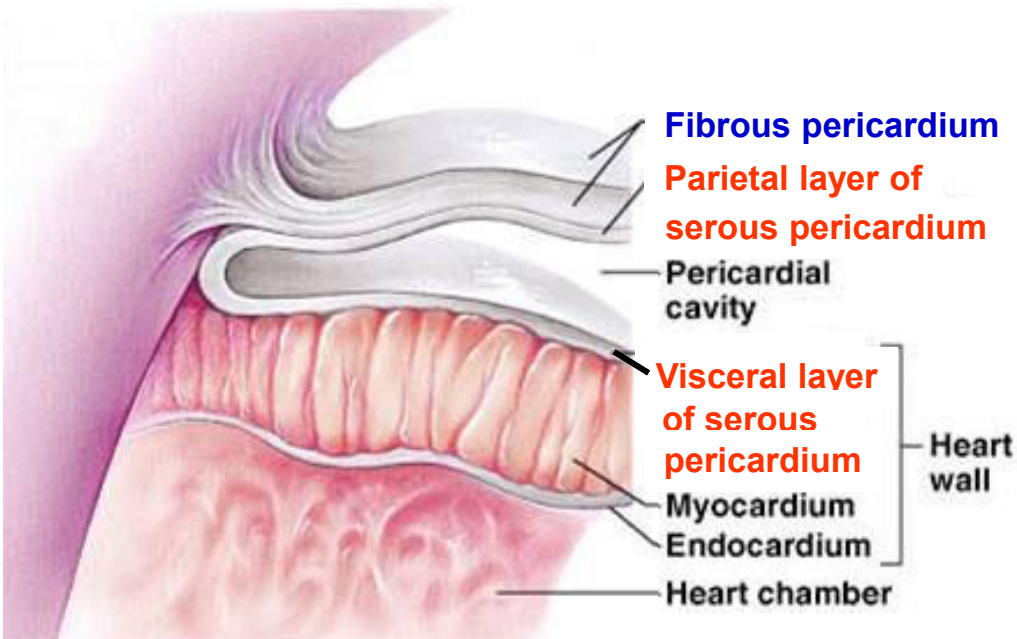


Pericardium



- Double walled sac around the heart & great vv. (proximal), composed of
 1. Superficial **fibrous pericardium**
 2. Deep two layers **serous pericardium** separated by pericardial cavity:

- **parietal layer**
- **visceral layer: epicardium**
- pericardial cavity: serous fluid**



Mesocardium

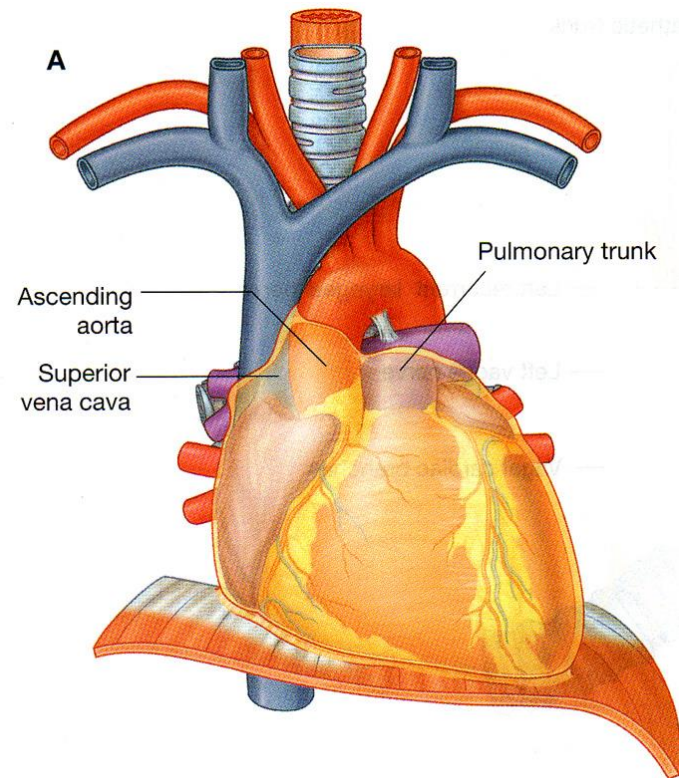
Epicardium covering the blood vessels which enter and leave the heart

1. Arterial mesocardium

- Ascending aorta
- Pulmonary trunk

2. Venous mesocardium

- Superior vena cava
- Lt. and Rt. pulmonary veins
- Inferior vena cava

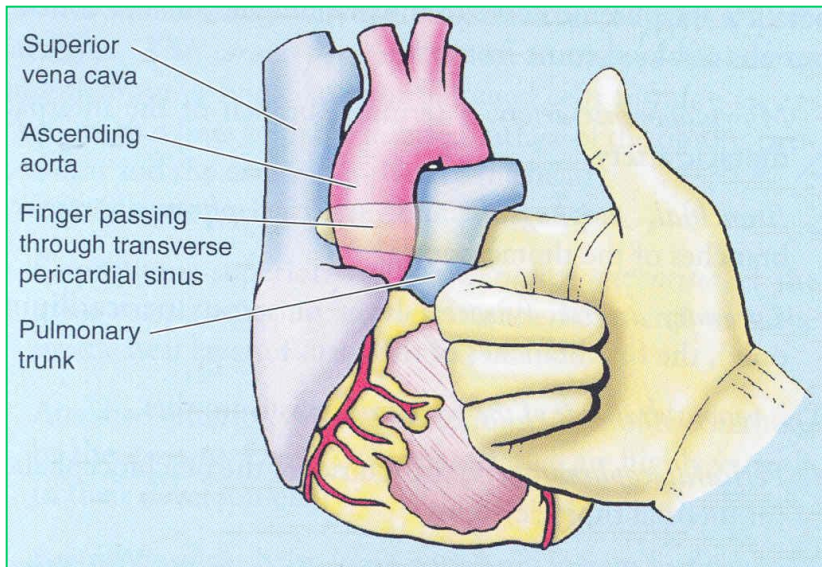


Pericardial sinuses

Pericardial reflections form the walls for sinuses in the pericardial space:

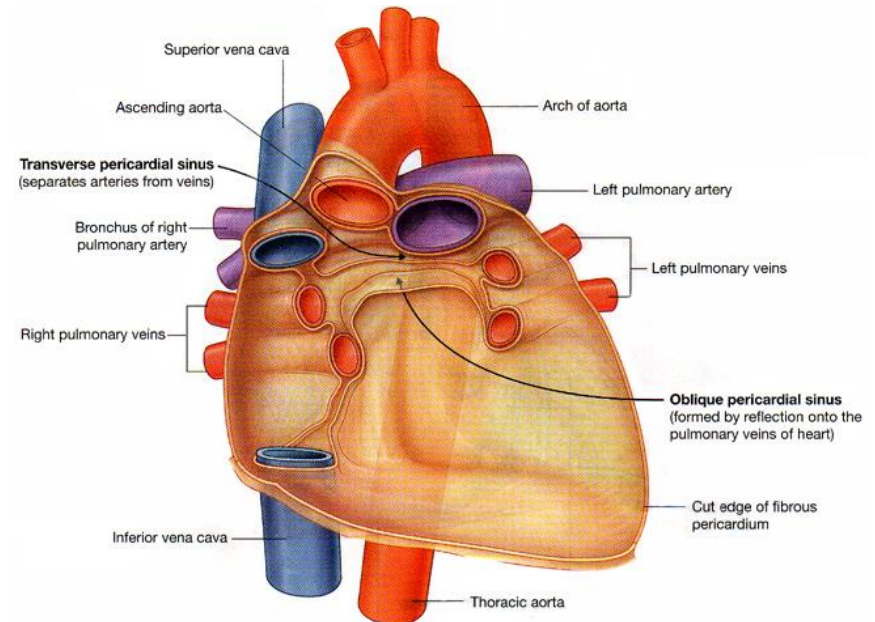
1. Transverse sinus

- a “tunnel” behind aorta and pulmonary trunk



2. Oblique sinus

- a “cul-de-sac” behind the left atrium formed by the parietal pericardium between pulmonary veins



Blood and nerve supplies of pericardium

Blood supply

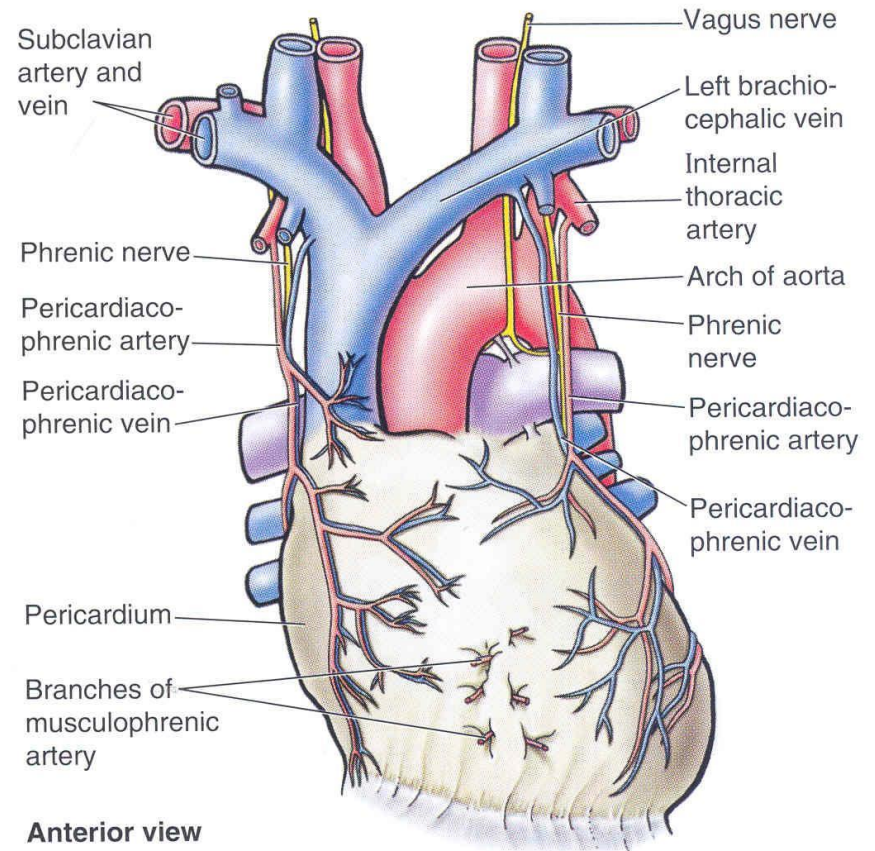
- Pericardiophrenic a. (mainly)
- Musculophrenic a.
- Coronary a. ----> visceral layer

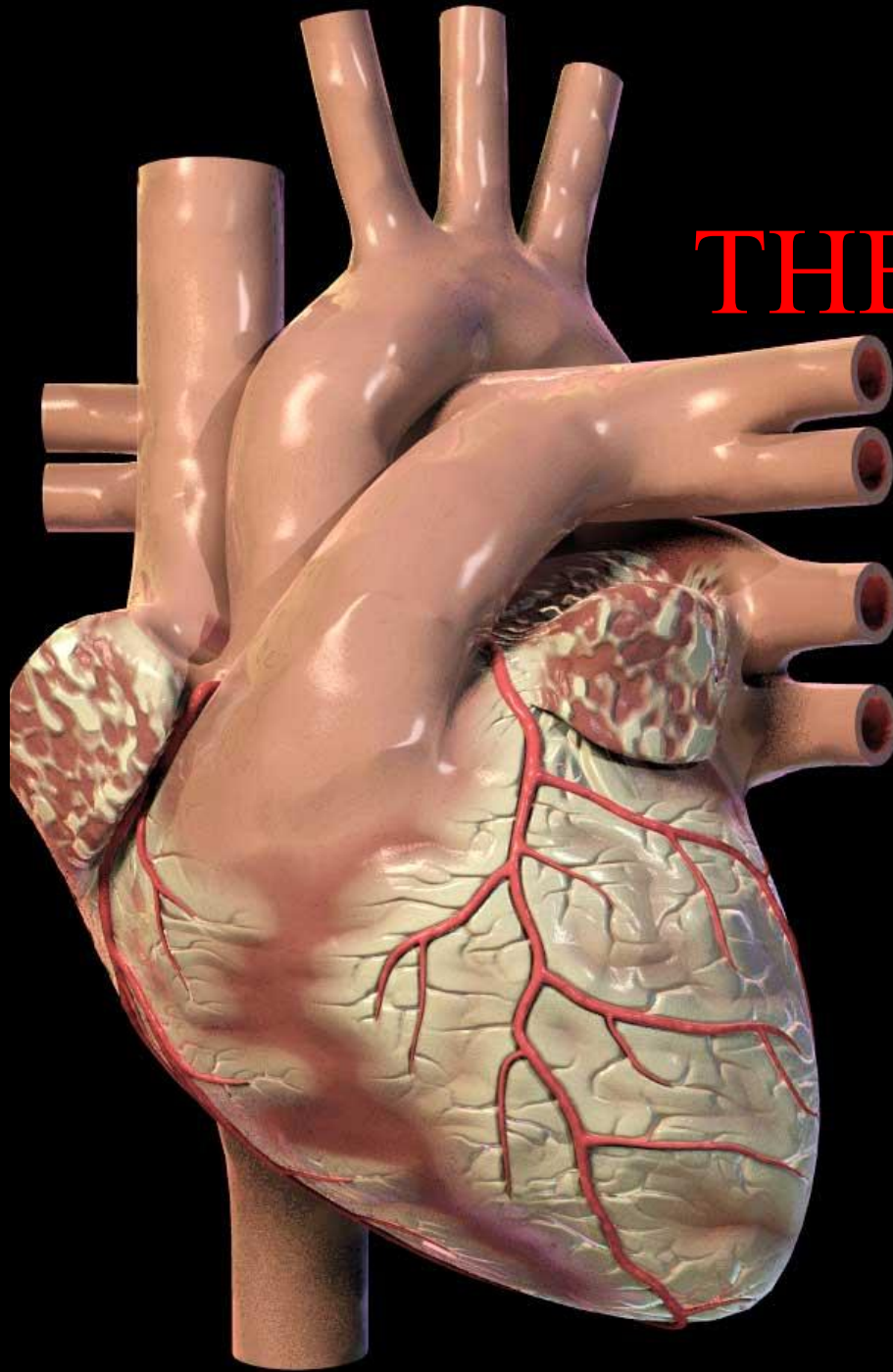
Venous drainage

- Pericardiophrenic v.

Nerve supply

- Phrenic n.





THE HEART

Heart

Base: Lt. atrium (mainly), Rt. Atrium

Apex: inferolateral part of Lt. ventricle
(5th Lt. intercostal space – midclavicular line)

Anterior (sternocostal) surface:

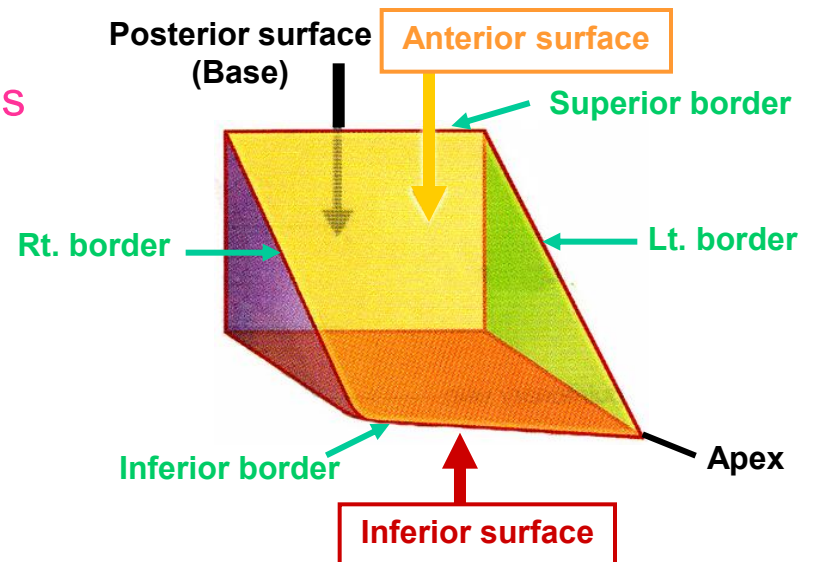
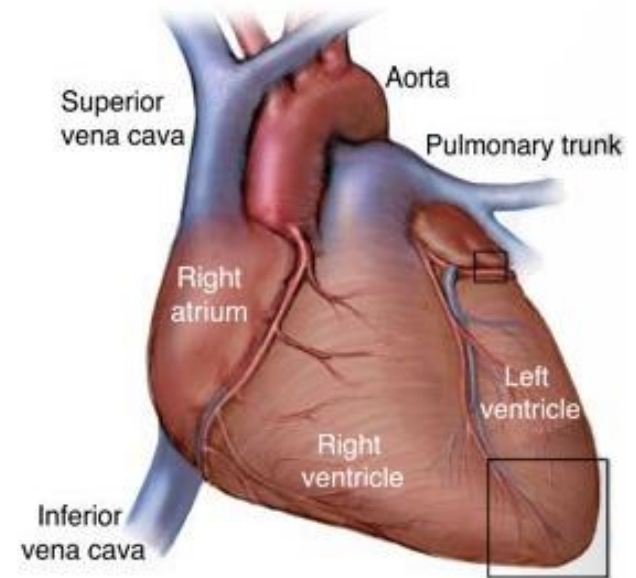
- Rt. atrium, Rt. ventricle, Lt. ventricle
- Anterior interventricular sulcus

Posterior surface

- Lt. atrium (mainly): 4 pulmonary veins
- Rt. Atrium
- Coronary (atrio-ventricular) sulcus

Inferior (diaphragmatic) surface

- Rt. Ventricle, Lt. ventricle
- Posterior interventricular sulcus



Heart

Base: Lt. atrium (mainly), Rt. Atrium

Apex: inferolateral part of Lt. ventricle

(5th Lt. intercostal space – midclavicular line)

Anterior (sternocostal) surface:

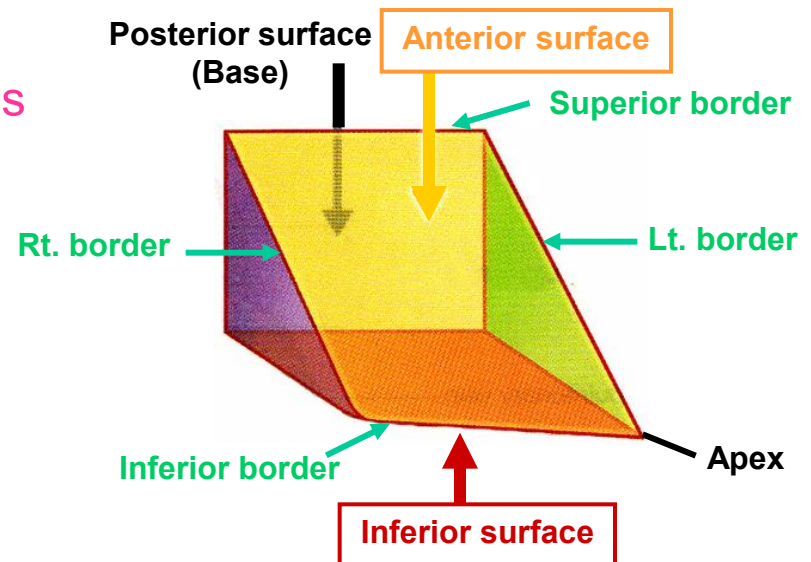
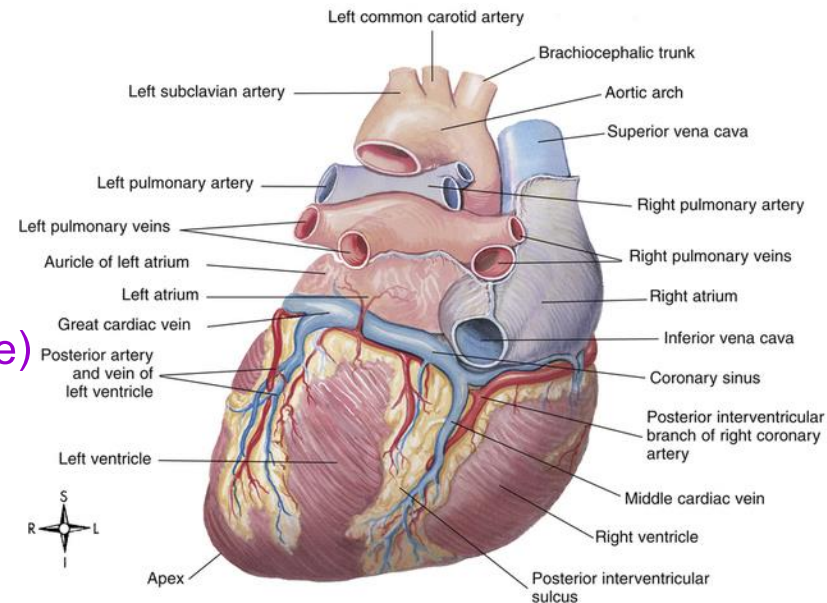
- Rt. atrium, Rt. ventricle, Lt. ventricle
- Anterior interventricular sulcus

Posterior surface

- Lt. atrium (mainly): 4 pulmonary veins
- Rt. Atrium
- Coronary (atrio-ventricular) sulcus

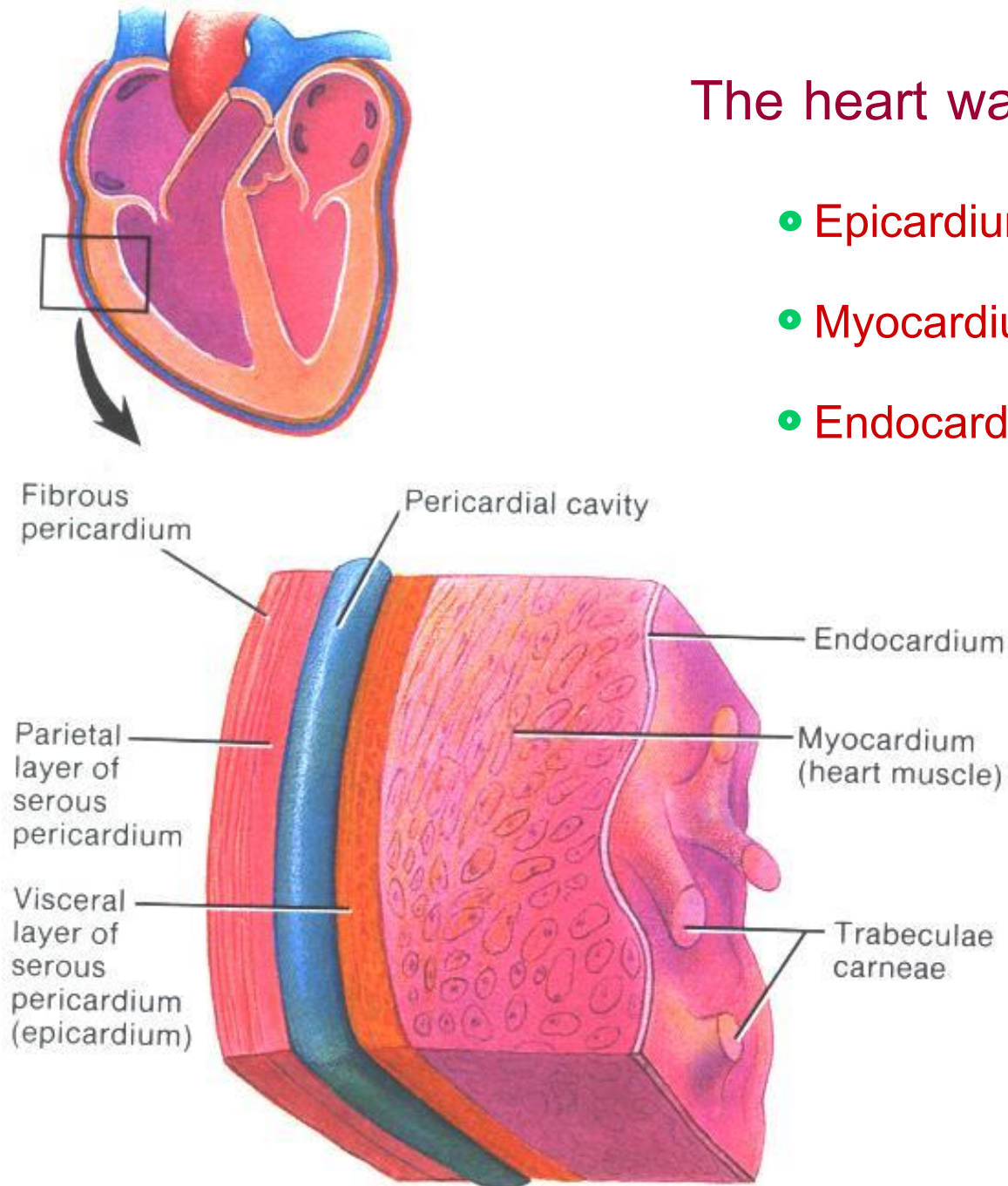
Inferior (diaphragmatic) surface

- Rt. Ventricle, Lt. ventricle
- Posterior interventricular sulcus



The heart wall

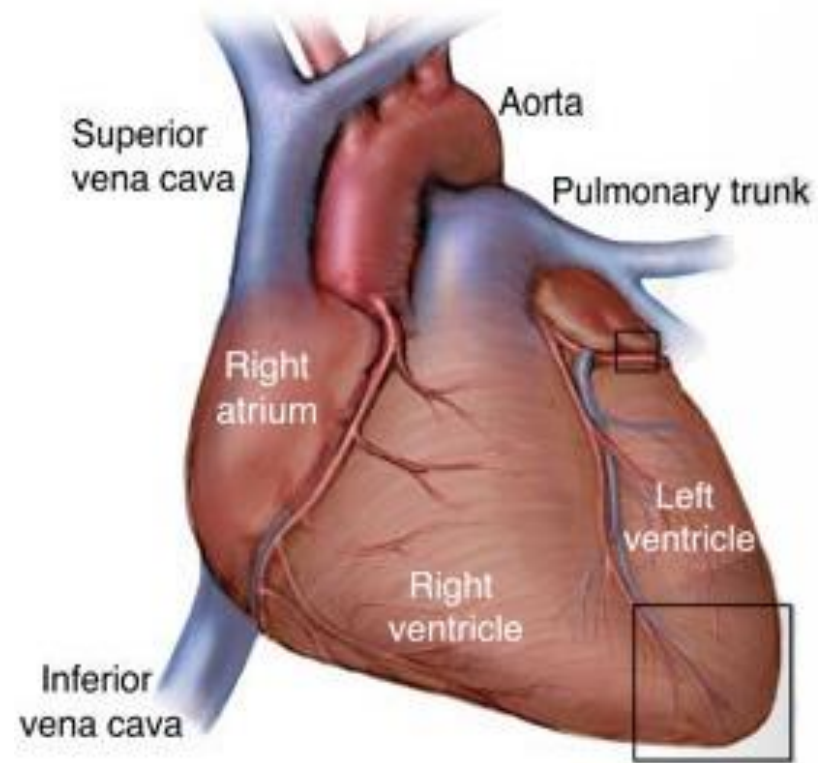
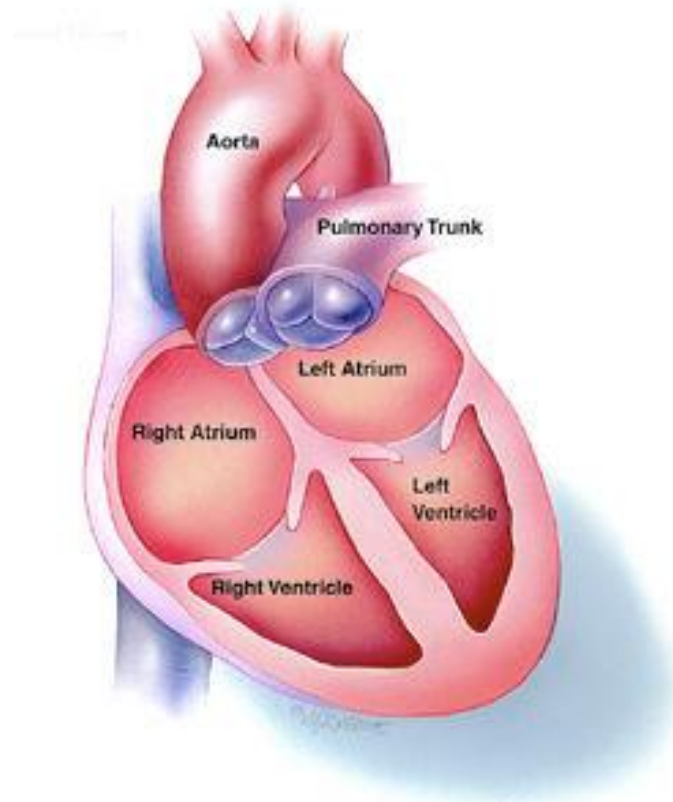
- Epicardium
- Myocardium
- Endocardium



Atria

Receiving chambers

Thin wall



Divided from each other by:

- interatrial septum

Divided from ventricles by

- atrio-ventricular sulcus
(coronary sulcus)

Right atrium

Receives deoxygenated blood

- superior vena cava
- inferior vena cava
- coronary sinus

Posterior wall

- sinus venarum (smooth)

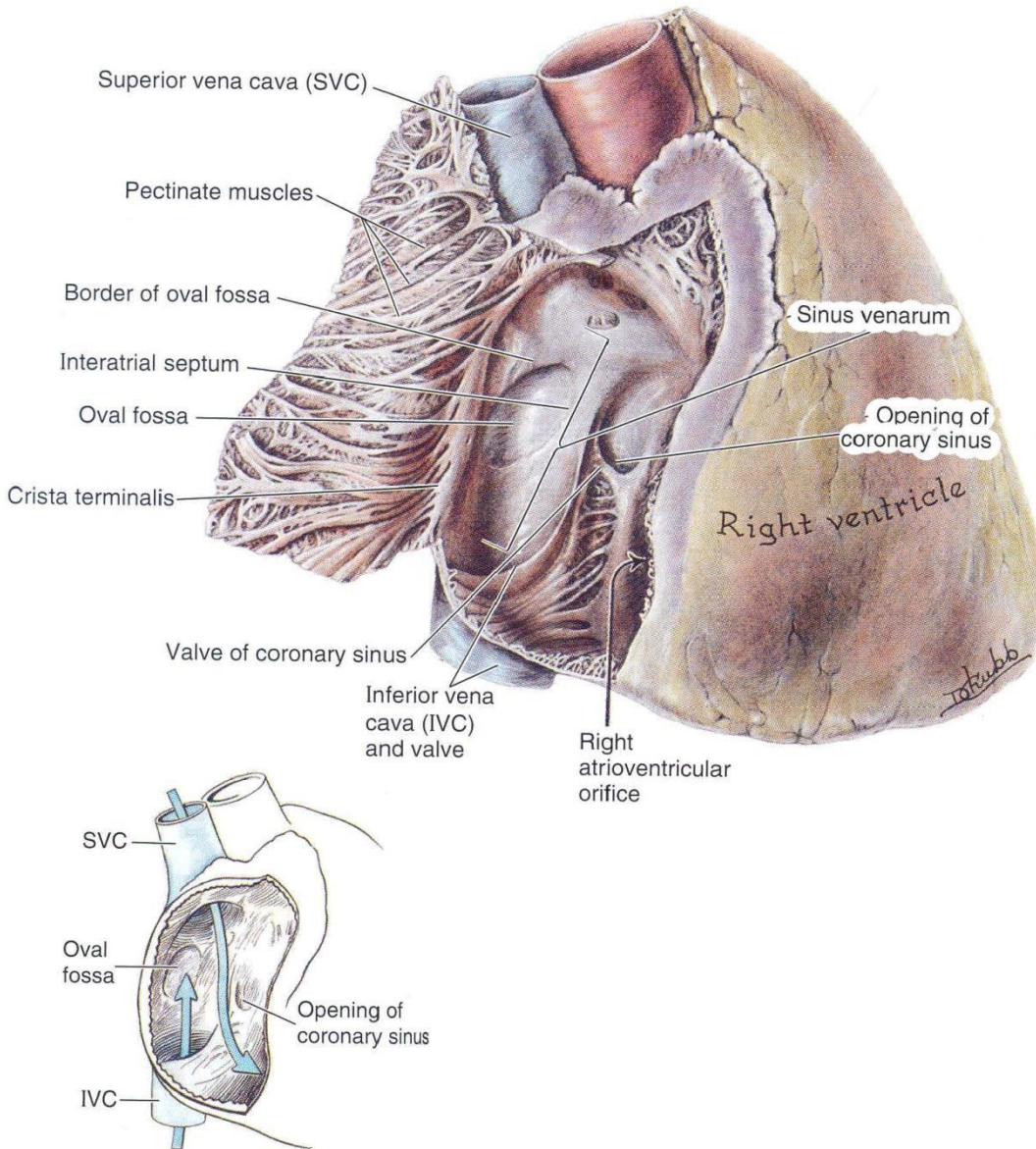
Anterior wall

- pectinate muscles (ridged)

Crista terminalis

Fossa ovalis

Opening of coronary sinus



Left atrium

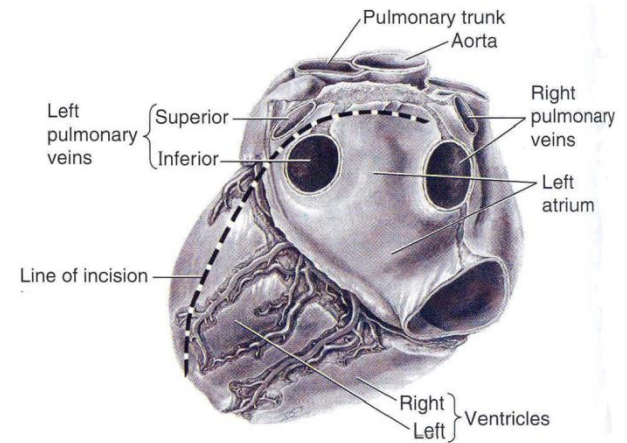
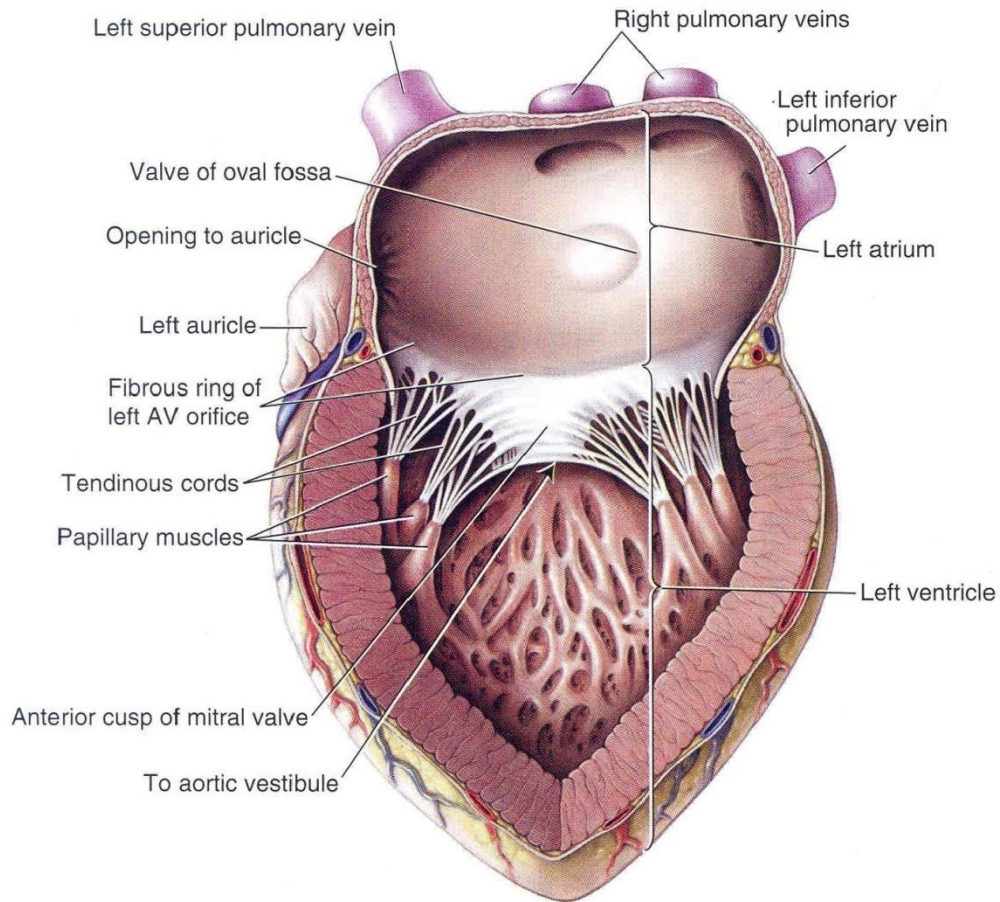
Form base

Receives oxygenated blood

- 2 right pulmonary veins
- 2 left pulmonary veins

Wall smooth

- except for pectinate muscles in auricle



Ventricles

Muscular pumping chambers

Separate externally :

- interventricular sulcus

Separate internally :

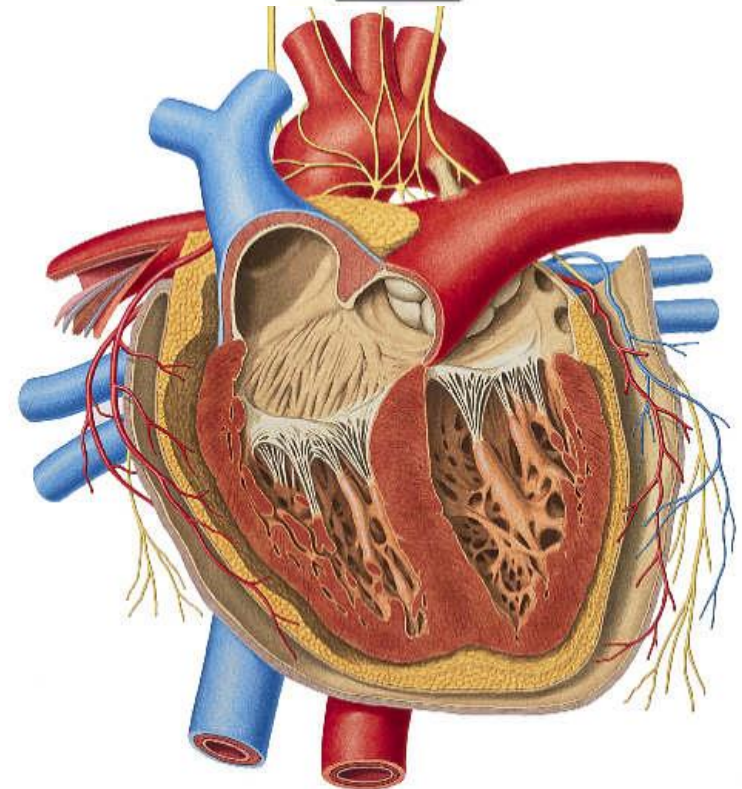
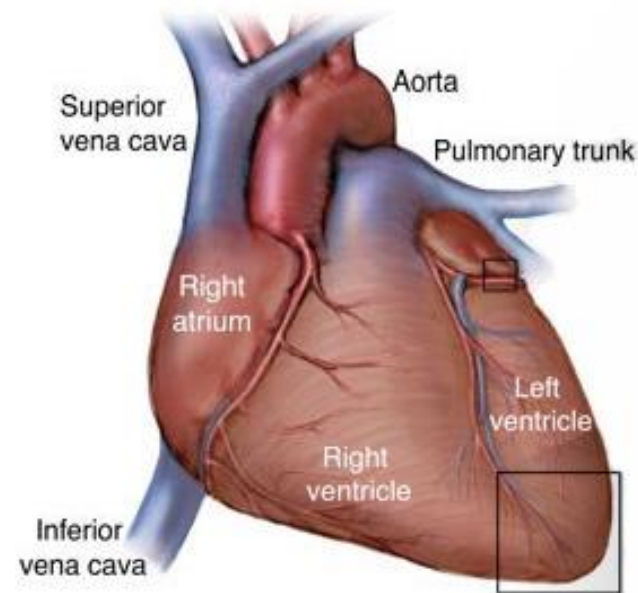
- interventricular septum

Trabeculae carnae

- muscle ridges

Papillary muscle

- cone-shaped muscle to which
chordae tendinae are anchored

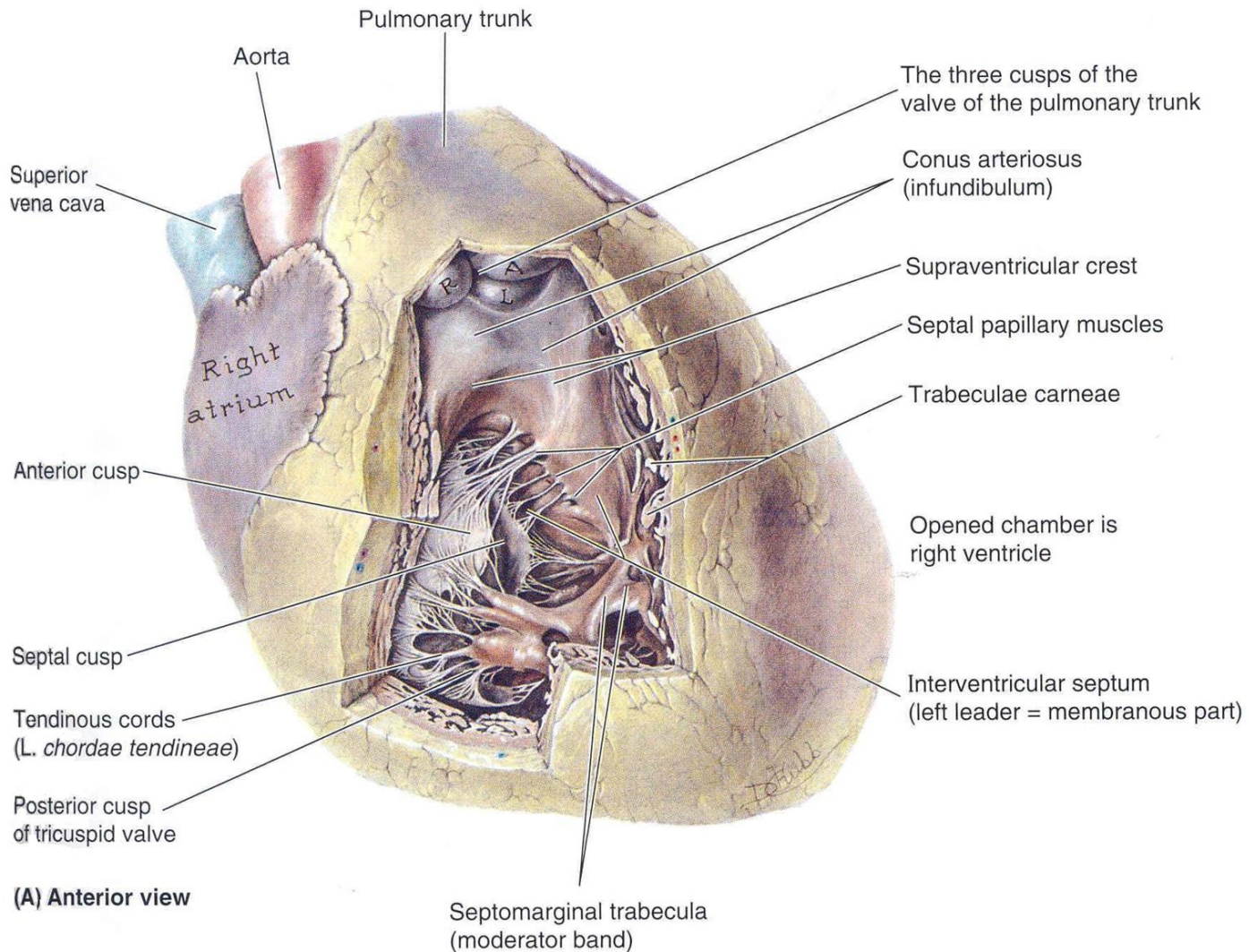


Right ventricle

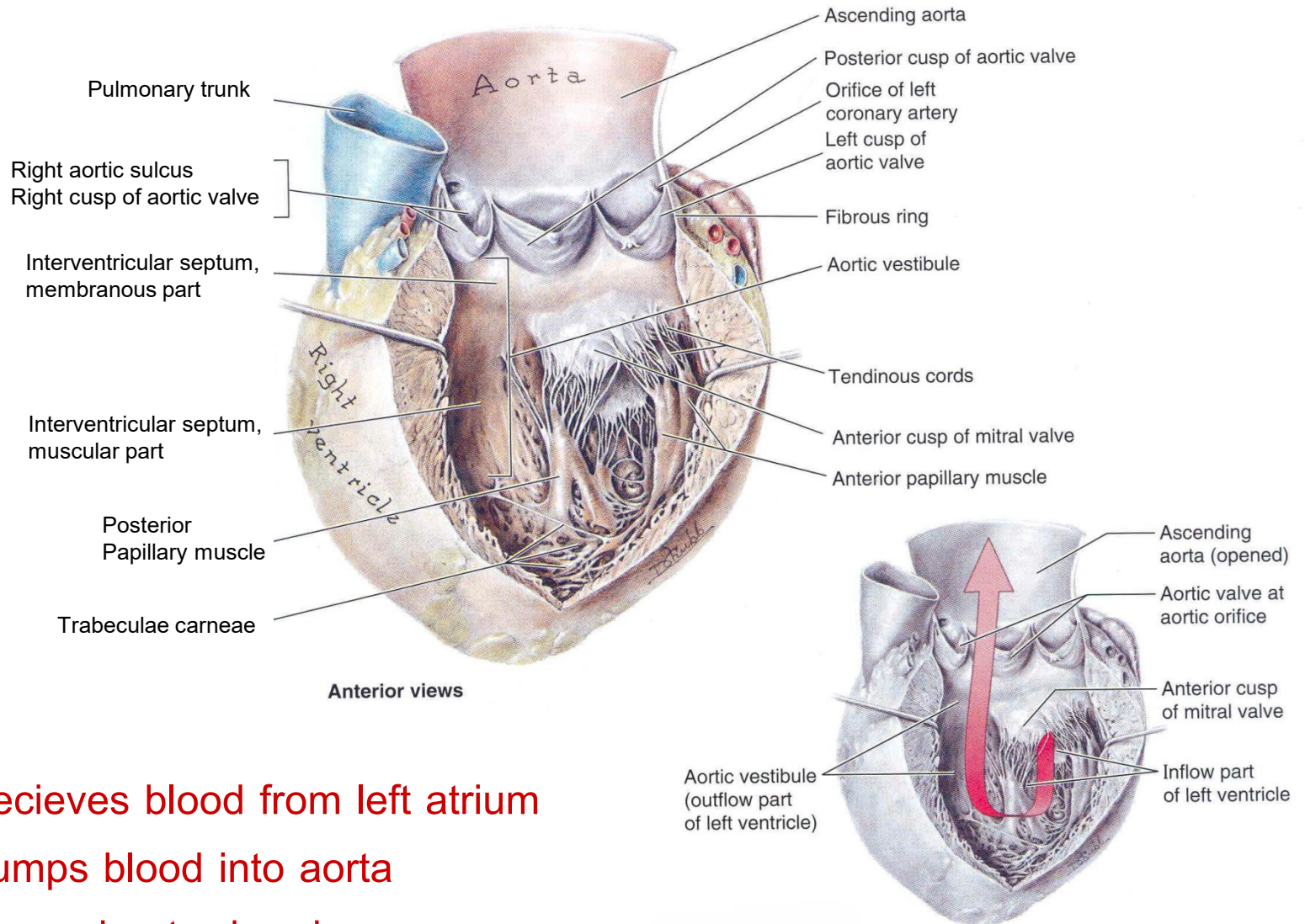
Receives blood from right atrium

Septomarginal (moderator) band: muscular band

- anterior papillary muscle to interventricular septum



Left ventricle



- Receives blood from left atrium
- Pumps blood into aorta
- No moderator band

Valves of the heart

Right

1. Tricuspid valve

: Rt. atrium – Rt. ventricle

2. Pulmonary valve

Rt. ventricle – pulmonary trunk

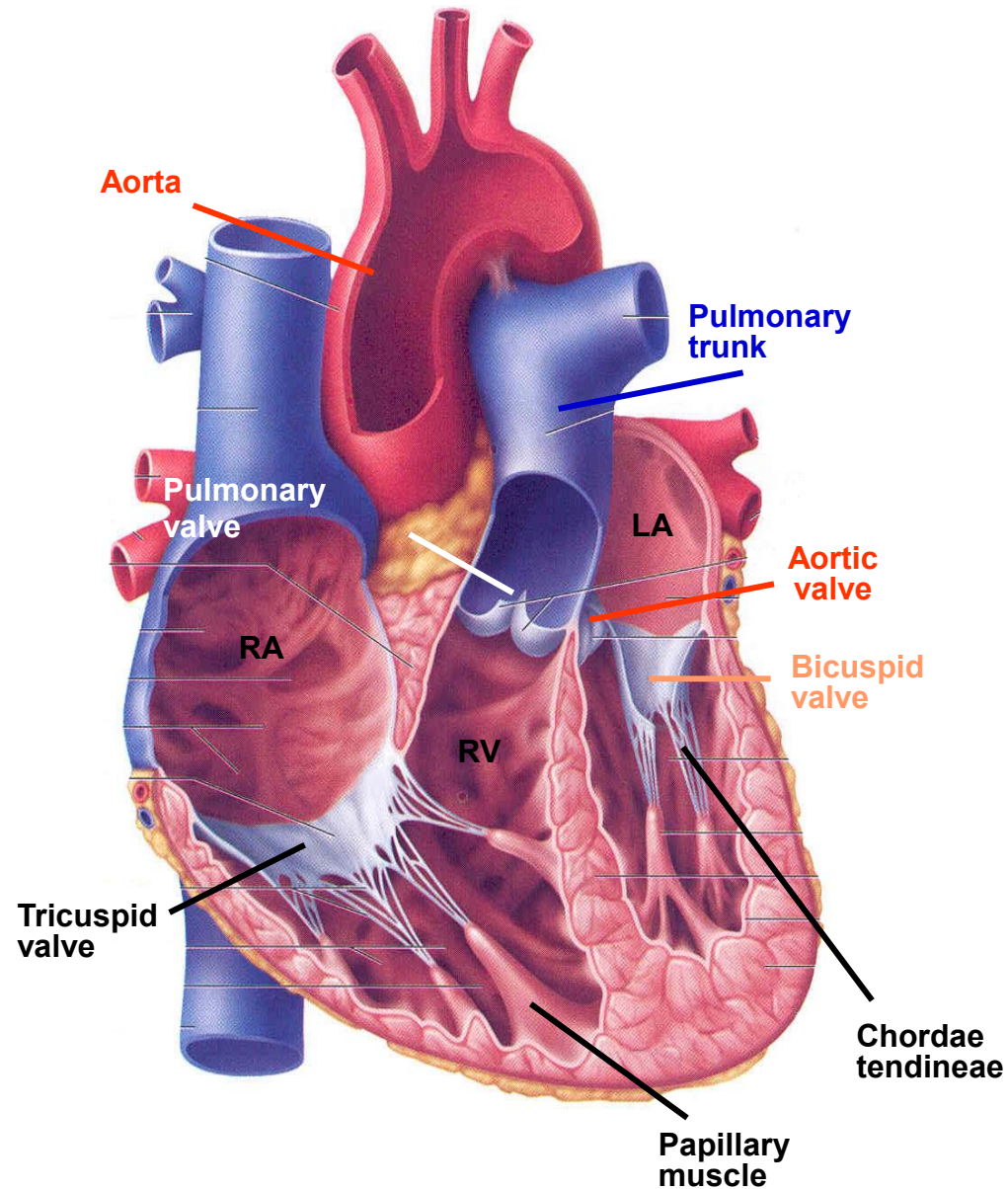
Left

1. Bicuspid (mitral) valve

: Lt. atrium – Lt. ventricle

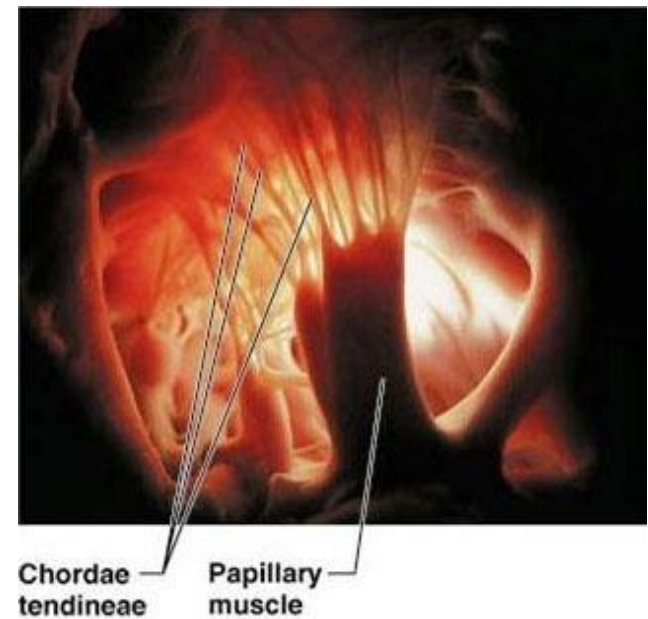
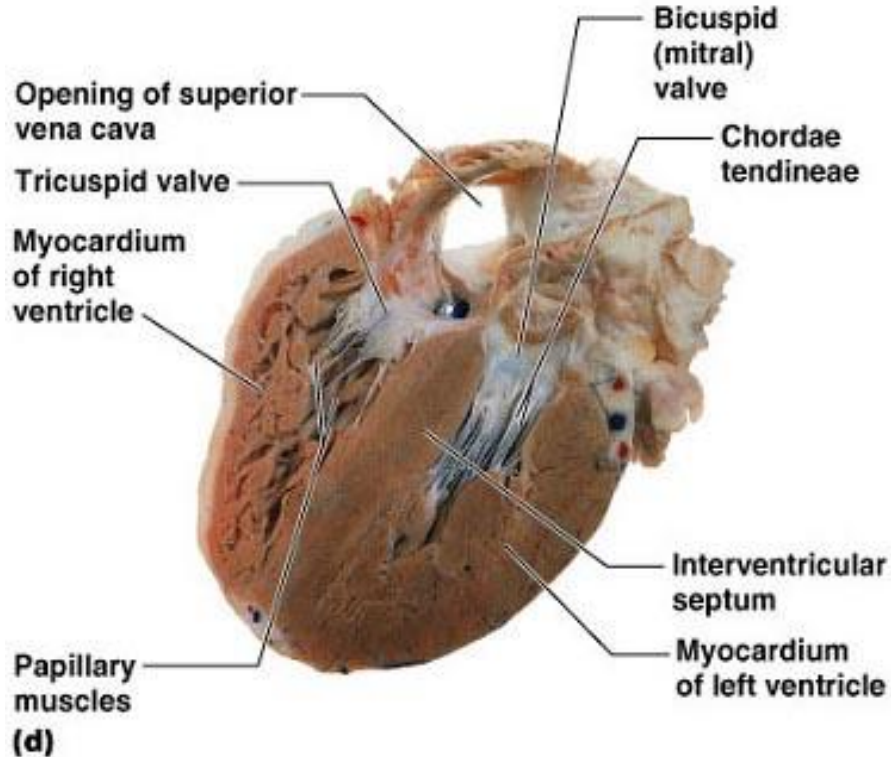
2. Aortic valve

: Lt. ventricle – aorta



Atrio-ventricular Valves

- Tricuspid valve
- Bicuspid (Mitral) valve



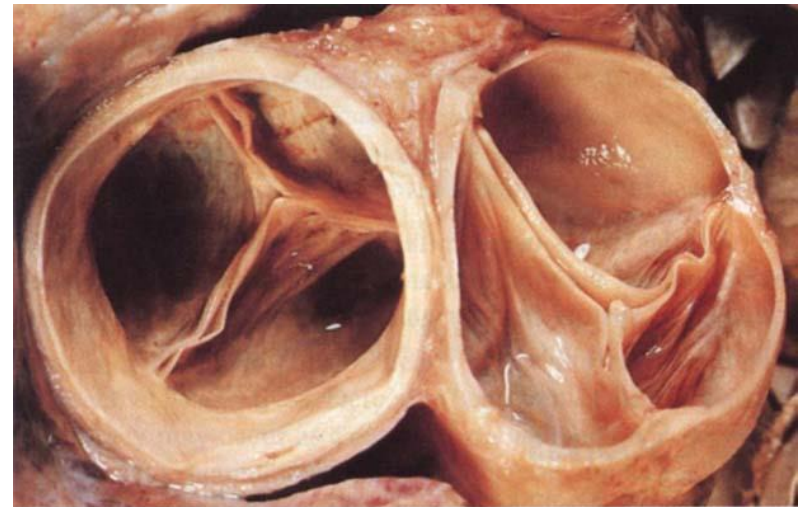
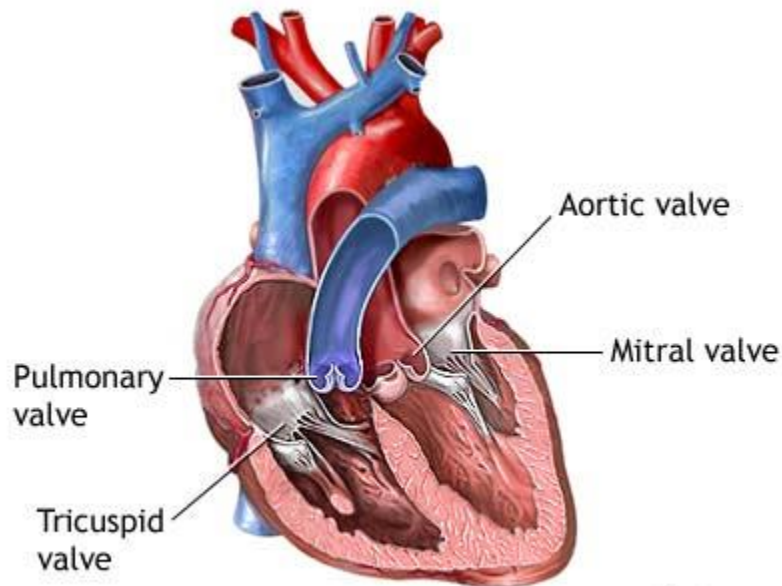
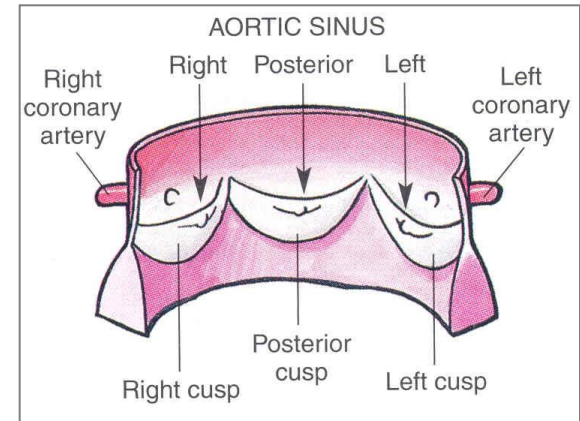
- structures

1. cusps
2. chordae tendinae
3. papillary muscles

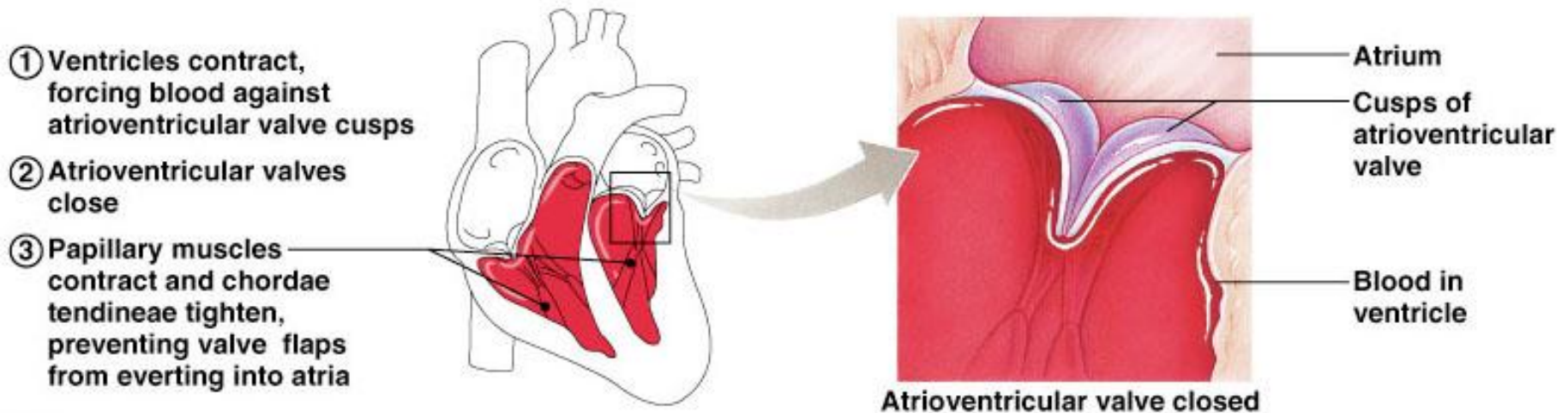
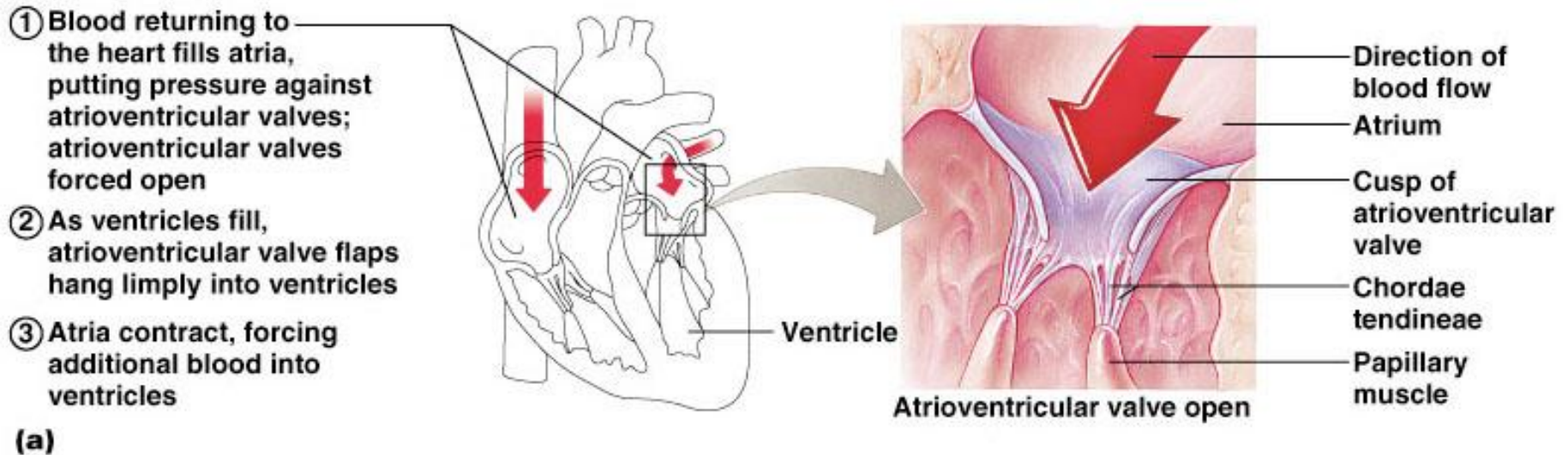
- prevent backflow into the atria when ventricles contract
- chordae tendineae anchor AV valves to papillary muscles

Semilunar Valves

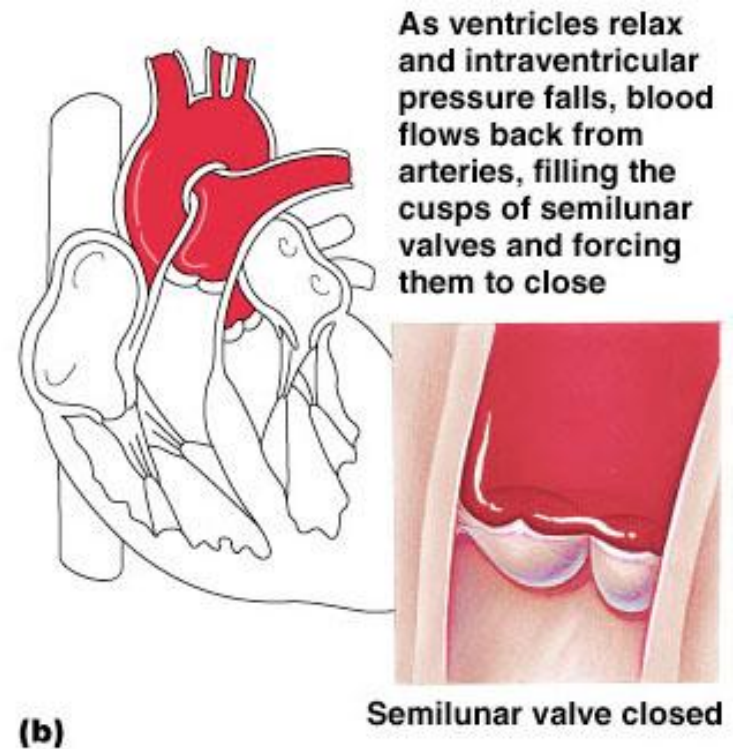
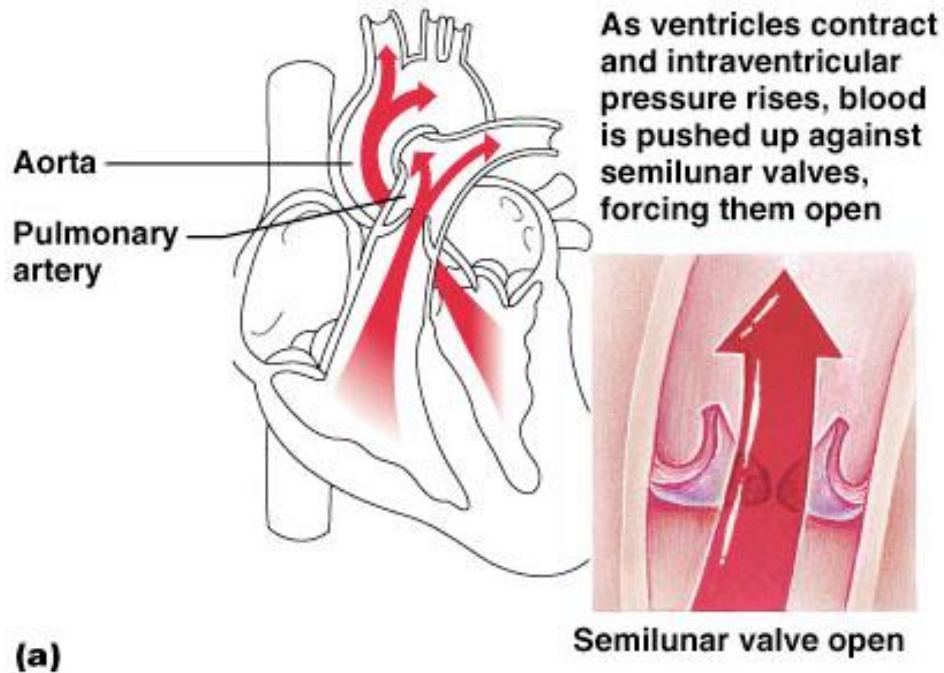
- Pulmonary valve
- Aortic valve
- 3 cusps
- open during ventricle contraction
- close when blood pushes back toward ventricle
- prevent backflow of blood into the ventricles



Atrioventricular Valve Function



Semilunar Valve Function



Auscultation of heart valves

Right 2nd ICS PSL



Auscultation position
for aortic valve

APT-M 2245 (Apartment-M 2245)

Left 2nd ICS PSL



Auscultation position
for pulmonary valve

Aortic valve

Pulmonary valve

Tricuspid valve

Mitral valve

Left 4th - 5th ICS PSL

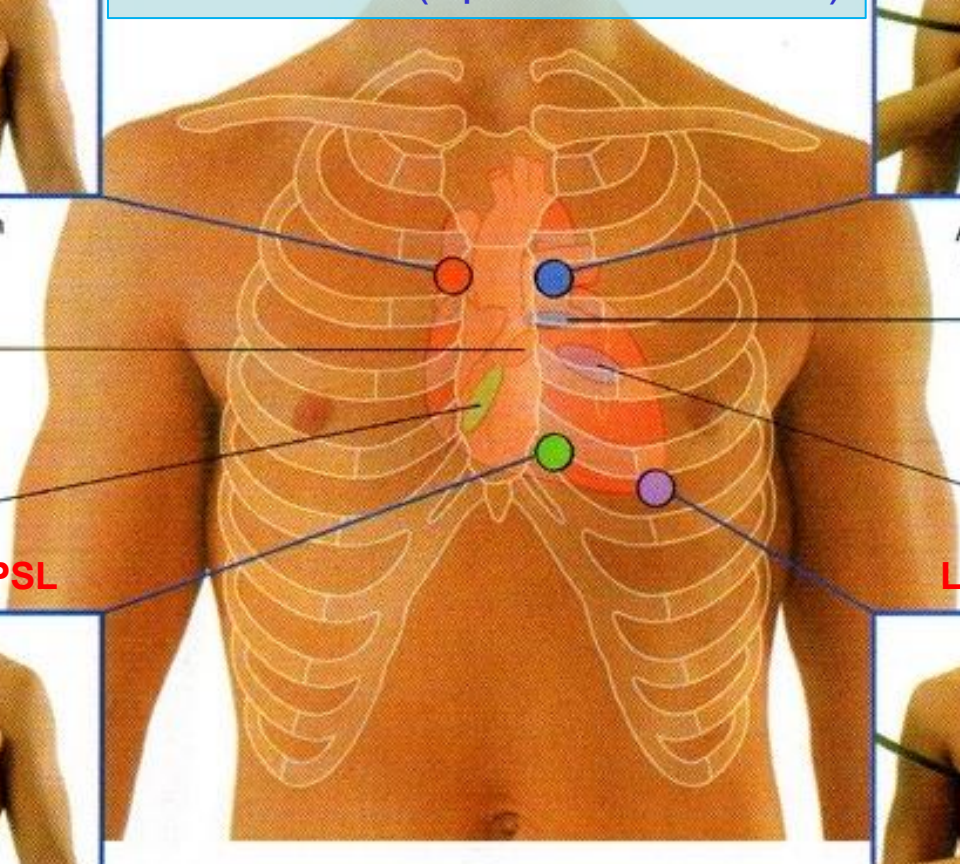


Auscultation position
for tricuspid valve

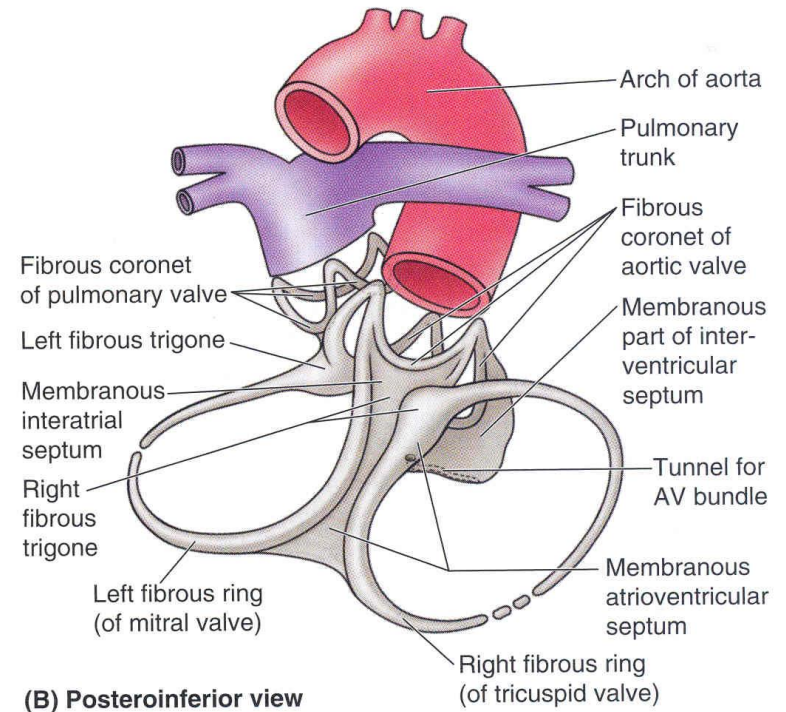
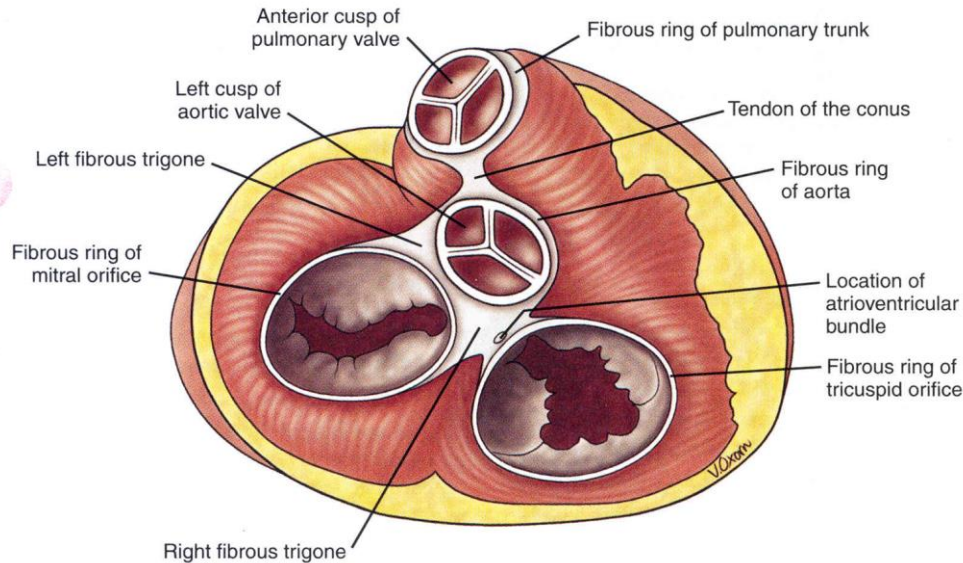
Left 5th ICS MCL



Auscultation position
for mitral valve

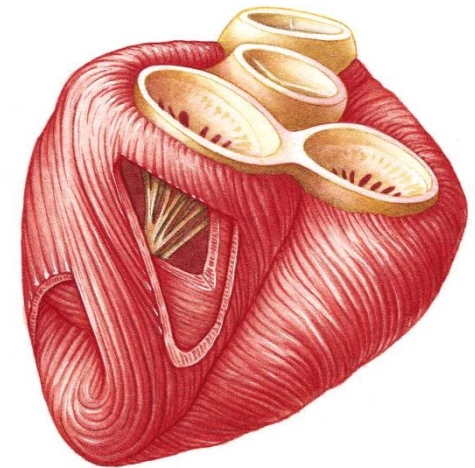


Cardiac skeleton

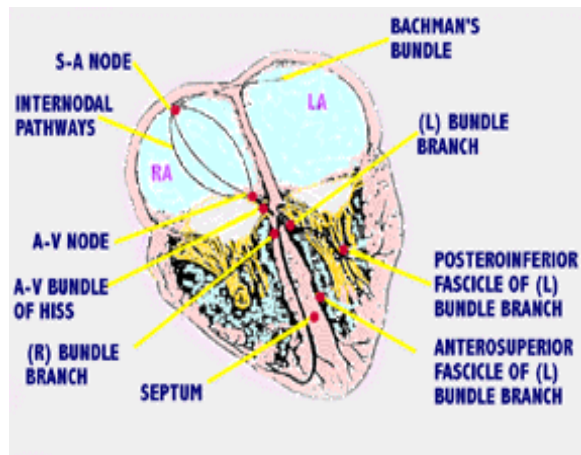
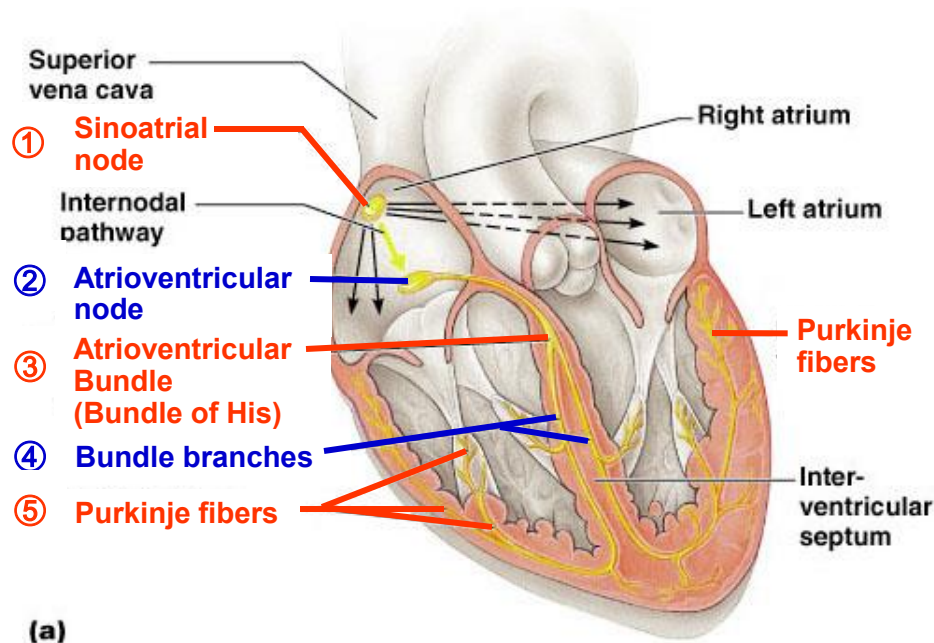


Fibrous tissue

- Atrioventricular orifice
- Root of aorta
- Root of pulmonary trunk
- Membranous part of interventricular septum



Conducting system of the heart



1. Sinoatrial (SA) node

- Located in right atrial wall just inferior to the opening of SVC
- Pacemaker that initiates each heartbeat

2. Atrioventricular (AV) node

- Located in the interatrial septum, just anterior to the opening of coronary sinus

3. Atrioventricular (AV) Bundle (Bundle of His)

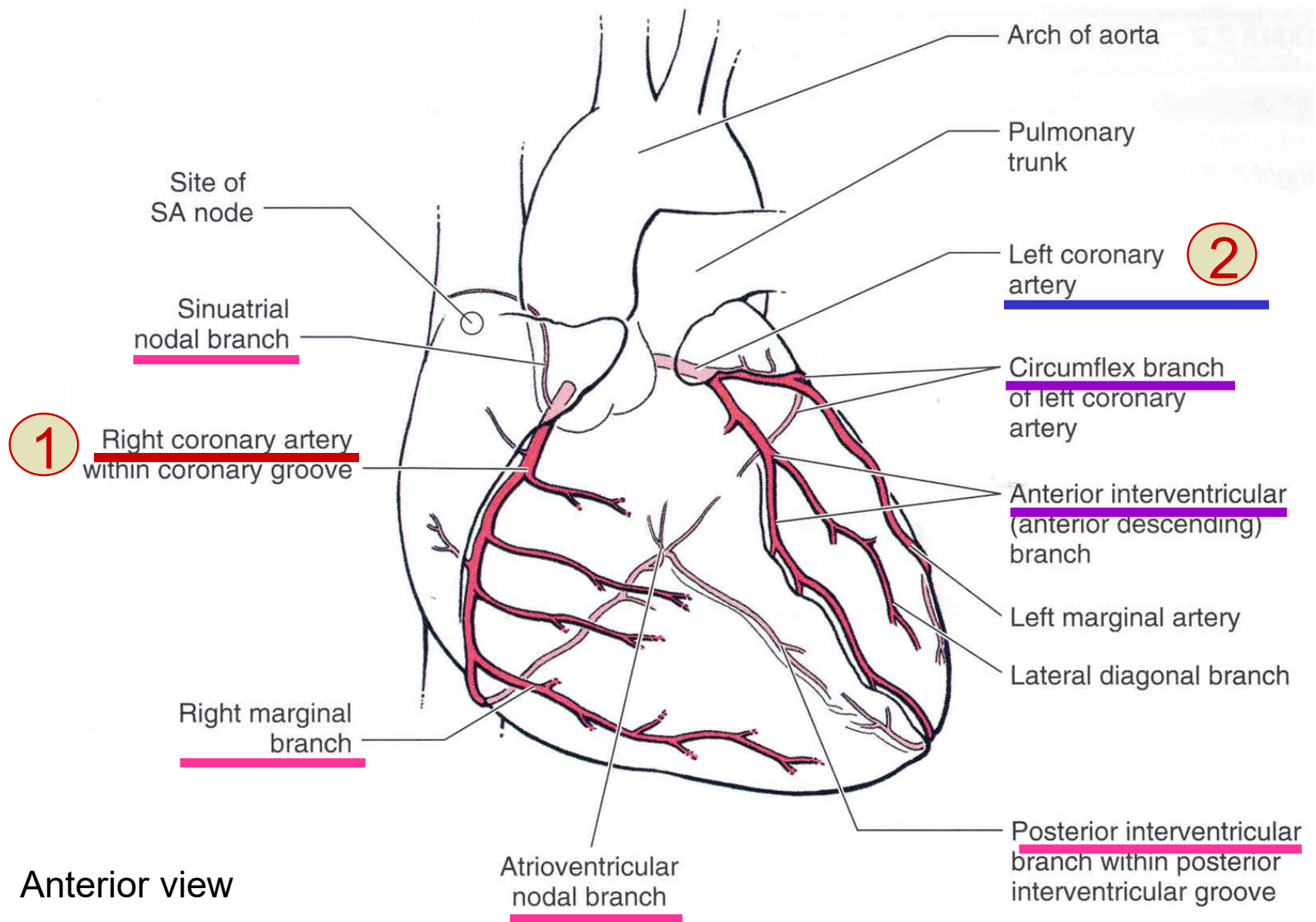
4. Bundles branches

5. Purkinje fibers

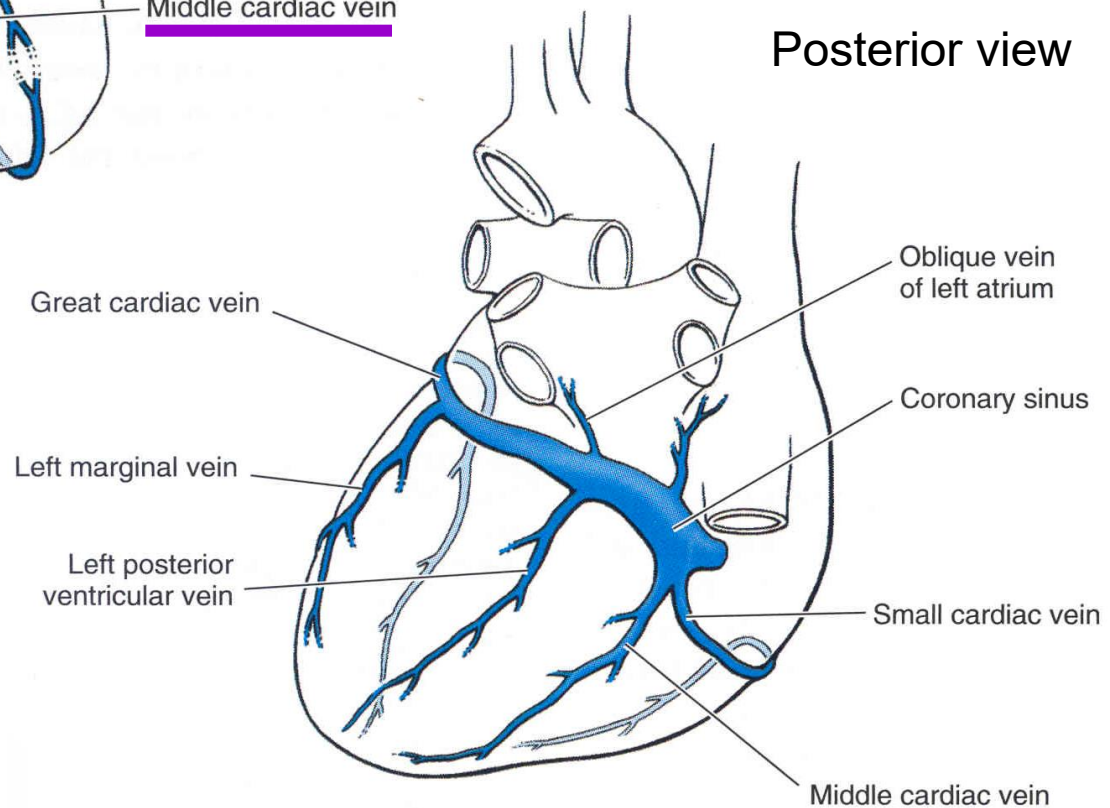
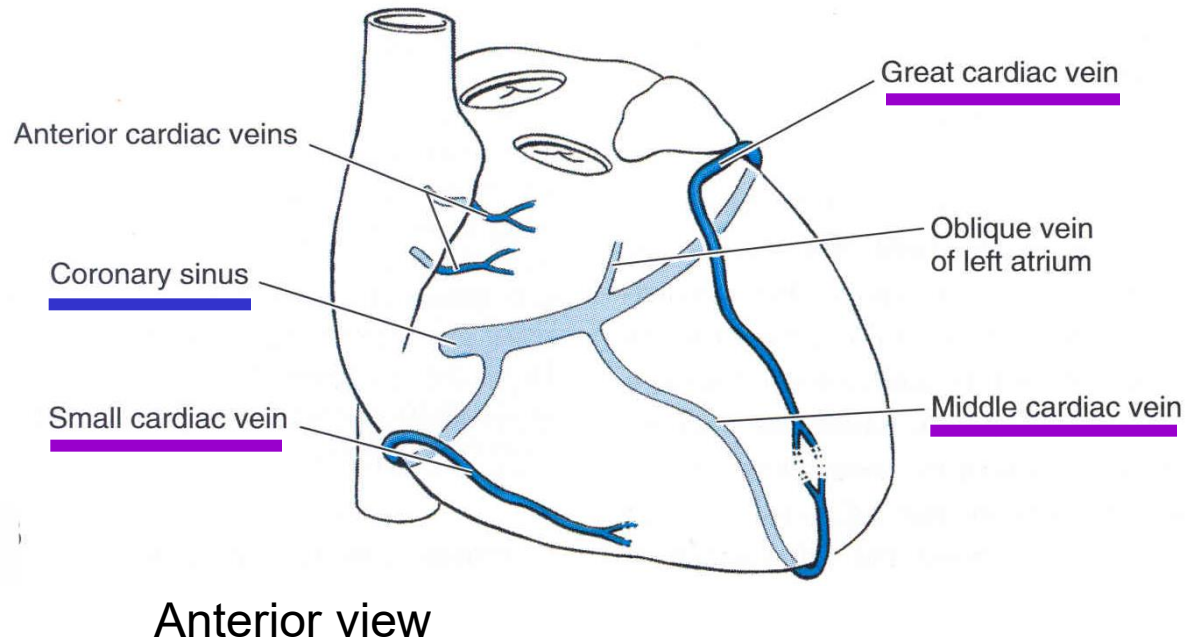


Coronary circulation

Arterial supply of the heart

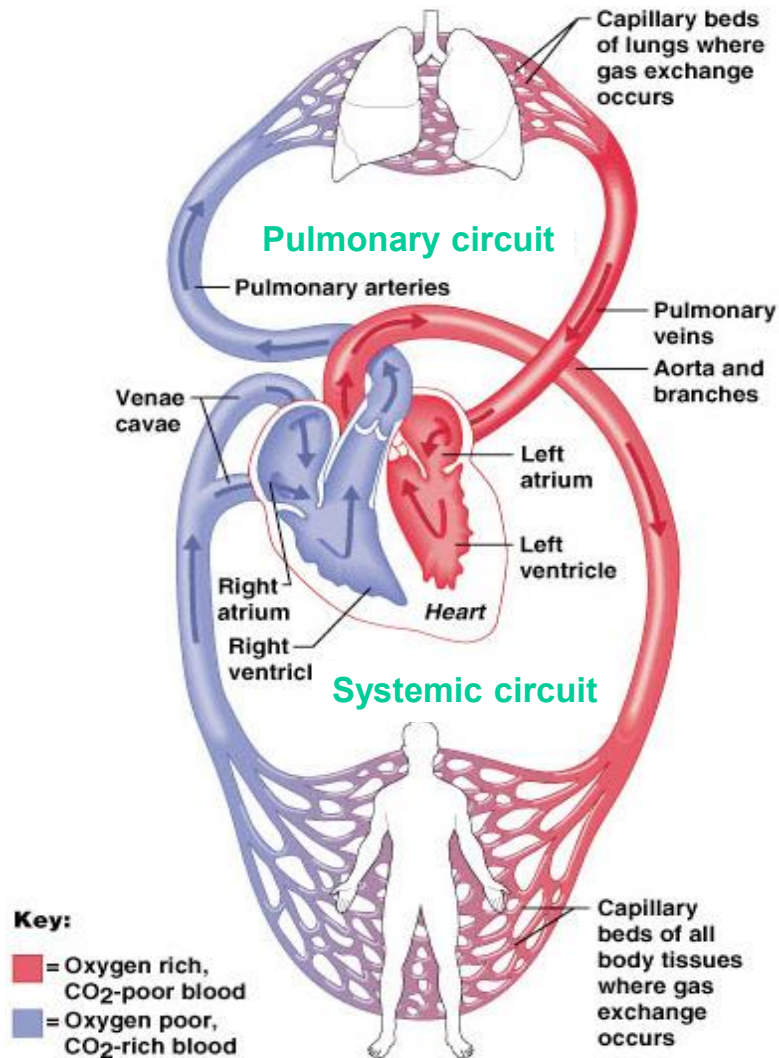


Venous drainage of the heart



Venae cardiac minimae

Circulatory Pathways



The vascular system has two distinct circulations

1. Pulmonary circuit

- Short loop that runs from the heart to the lungs and back to the heart

2. Systemic circuit

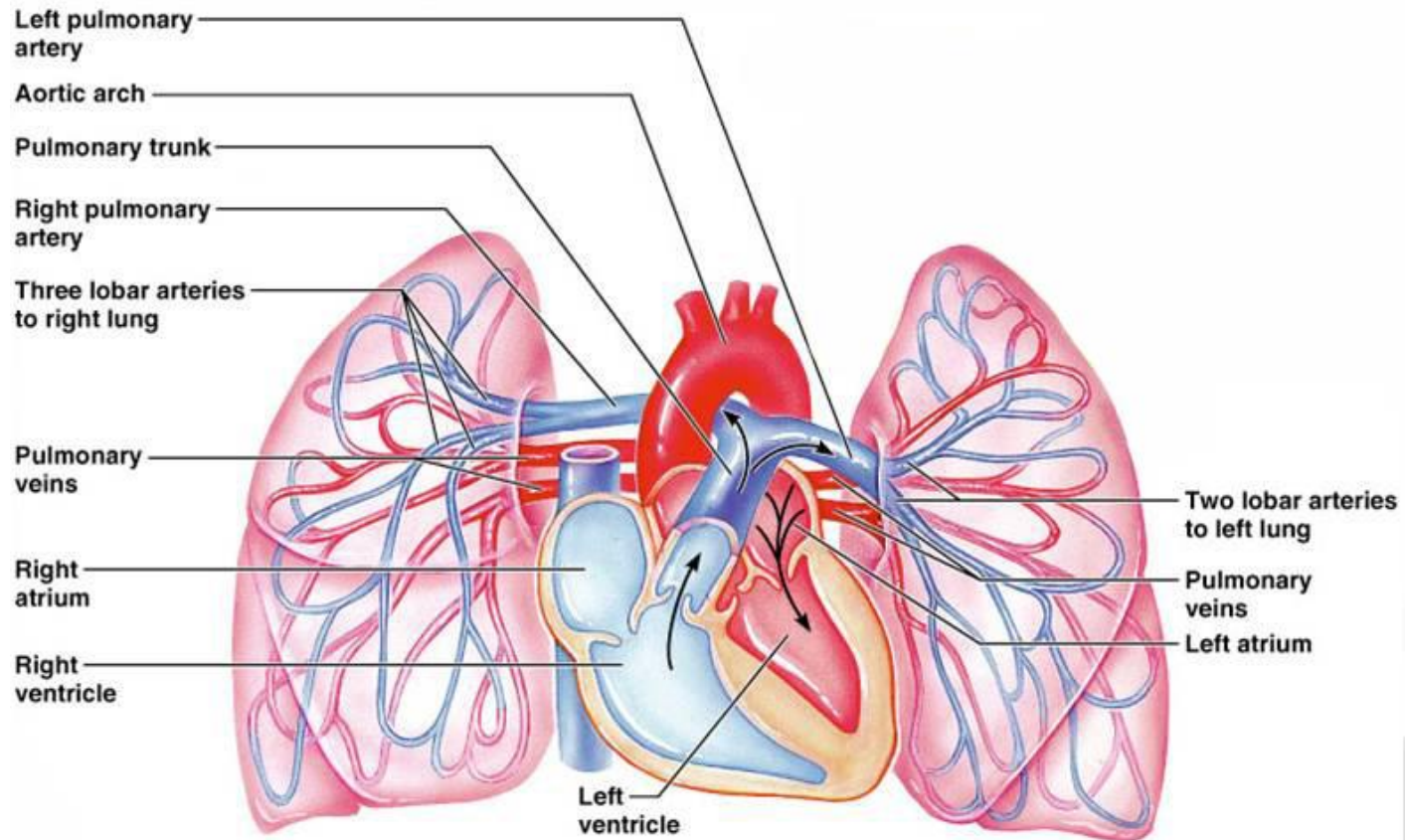
- Routes blood through a long loop to all parts of the body and returns to the heart

Pulmonary trunk

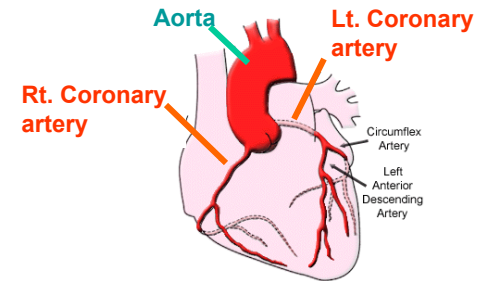
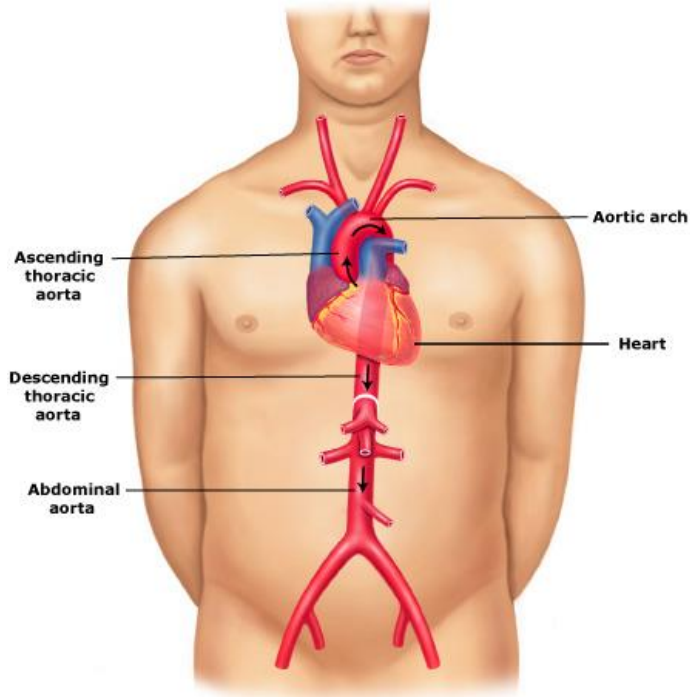
1. Right pulmonary artery
2. Left pulmonary artery

Pulmonary veins

1. Right pulmonary veins
2. Left pulmonary veins



Aorta

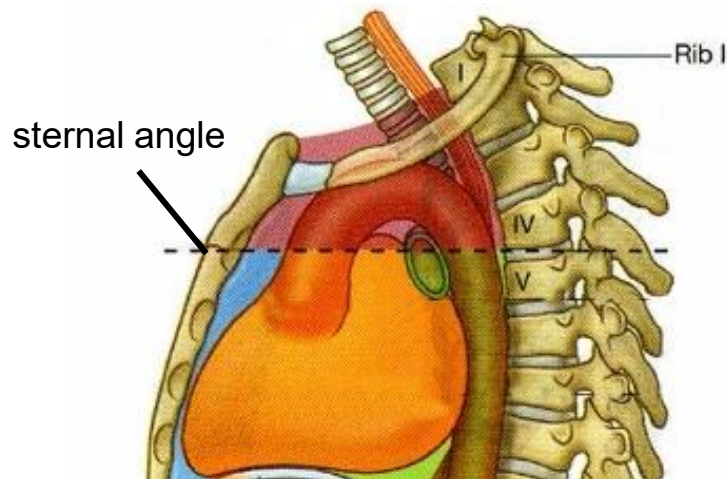


1. Ascending aorta

- Arises from LV: 2nd Lt. costal cartilage
- Branches: Rt., Lt. coronary a.

2. Arch of aorta

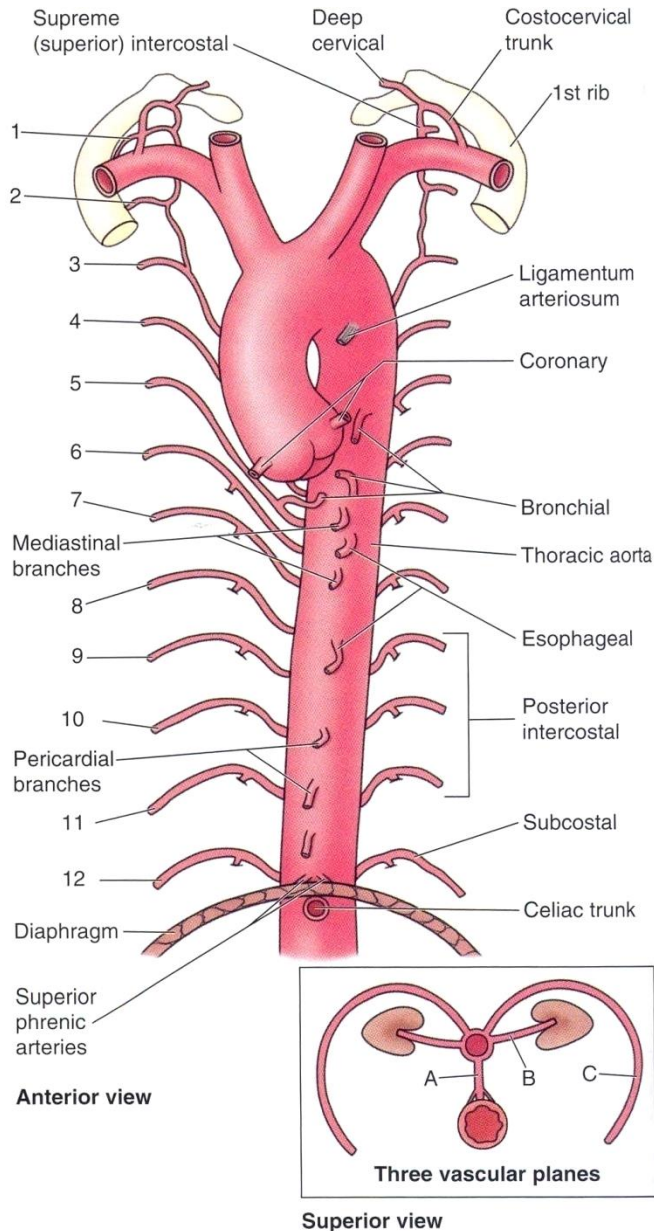
- Sternal angle – T4 vertebra
- Branches:
 - brachiocephalic trunk
 - Lt. common carotid a.
 - Lt. subclavian a.



3. Descending aorta

- Thoracic aorta
- Abdominal aorta: T12 vertebra

Thoracic aorta



1. Visceral branches

- Pericardial branches
- 2 Lt. Bronchial a.
(1 Rt. Bronchial a.: 3rd Rt. post^r intercostal a.)
- 2 esophageal a.

2. Parietal branches

- posterior intercostal a.
(ISC 3 -11)
- 1 pairs of subcostal a. (ICS 12)
- Superior phrenic a.
- Mediastinal branches

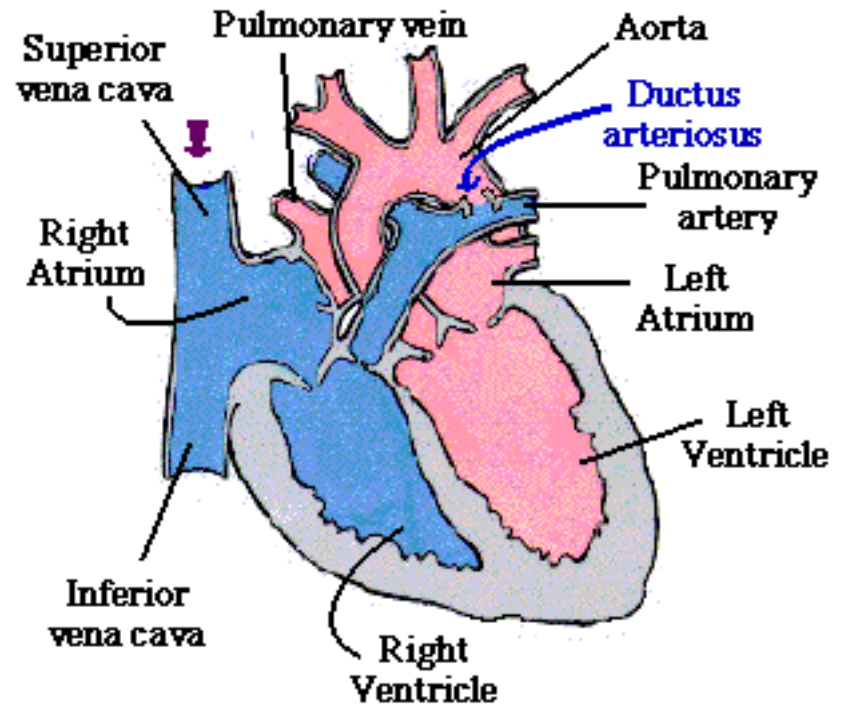
Ligamentum arteriosum

- Ductus arteriosus
- Left pulmonary a.



Inf^r surface of aortic arch
(Lt. subclavian a.)

- Lt recurrent laryngeal n.

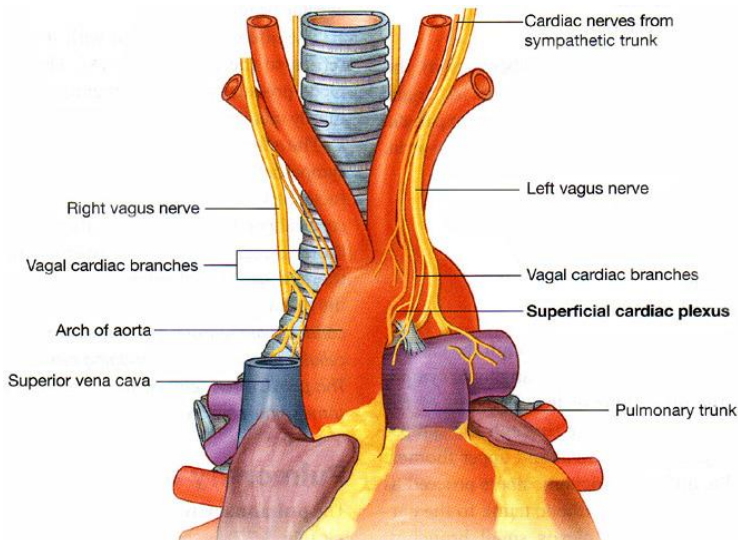


Structures below arch of aorta

Medial Superficial cardiac plexus



lateral Lt. recurrent laryngeal n.



Large veins of the thorax

1. Brachiocephalic veins: internal jugular v. + subclavian v.

Rt. Brachiocephalic v.

(shorter, more vertical)

- Rt. vertebral v.
- Rt. int. thoracic v.
- 1st Rt. post^r intercostal v.

Lt. Brachiocephalic v.

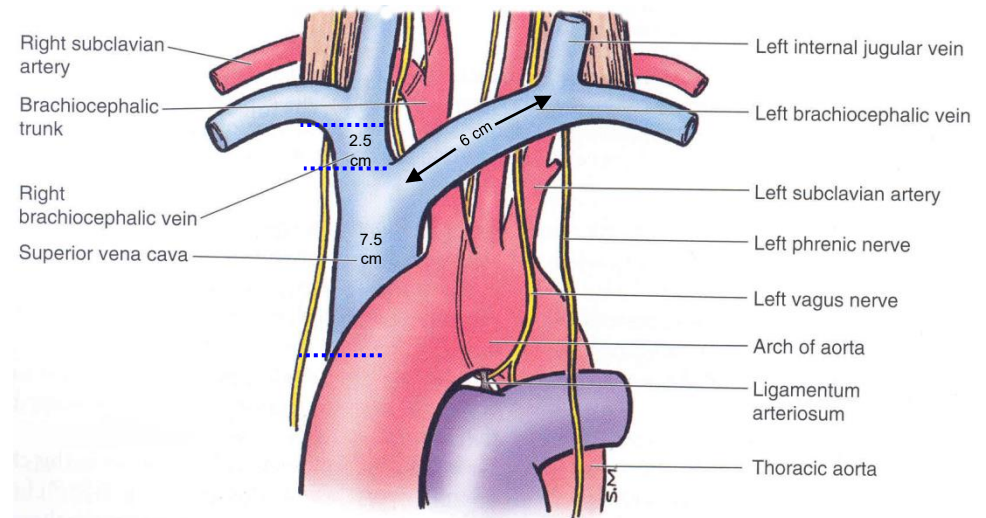
(longer, more horizontal)

- Lt. vertebral v.
- Lt. int. thoracic v.
- 1st Lt. post^r intercostal v.
- Inferior thyroid v., thymic v.

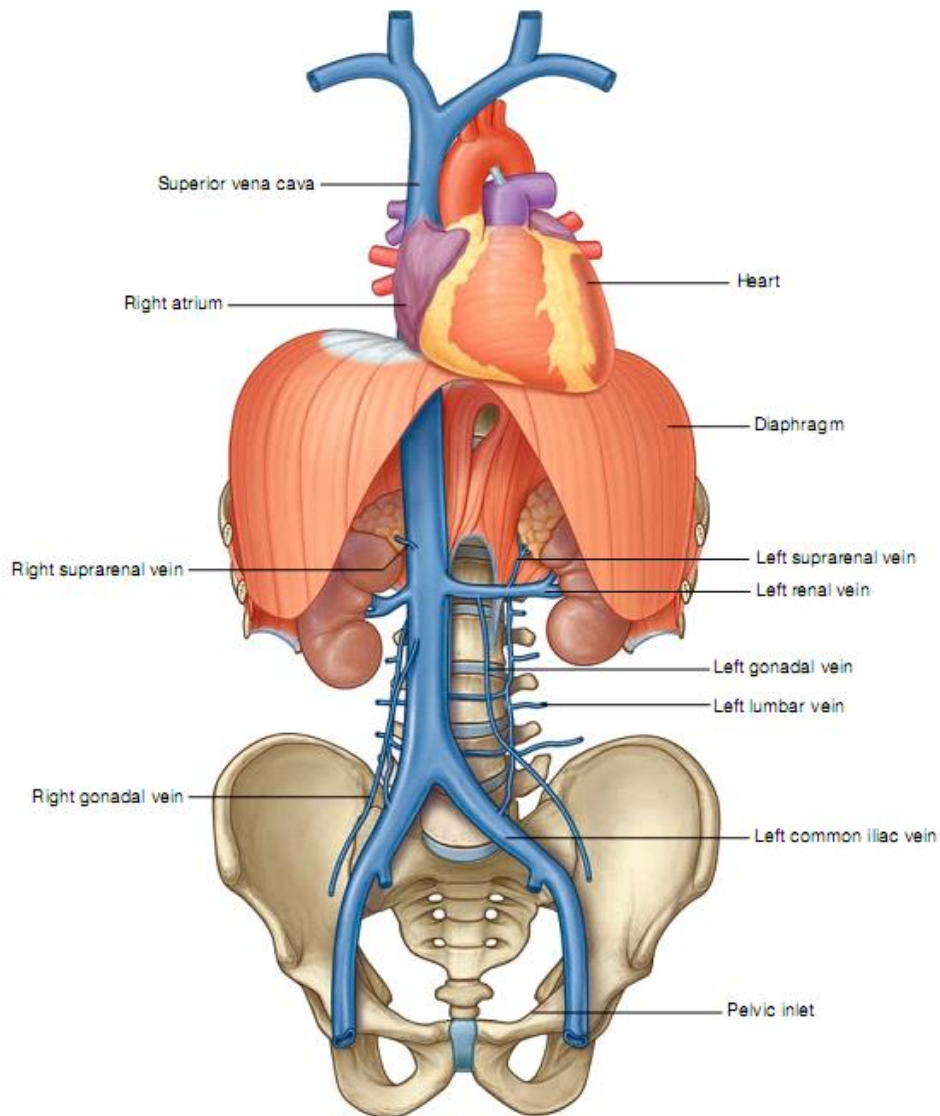
2. Superior vena cava (1st right costal cartilage)

- Azygos vein

Right atrium
(3rd right costal cartilage)



Inferior vena cava



Right atrium



Caval opening
(T8 vertebra)



Rt. + Lt. common iliac veins

Azygos system of veins

Posterior intercostal veins

1 pair: subcostal veins

11 pairs: posterior intercostal veins

- 1st: brachiocephalic vein
- 2nd - 4th ---> superior intercostal vein

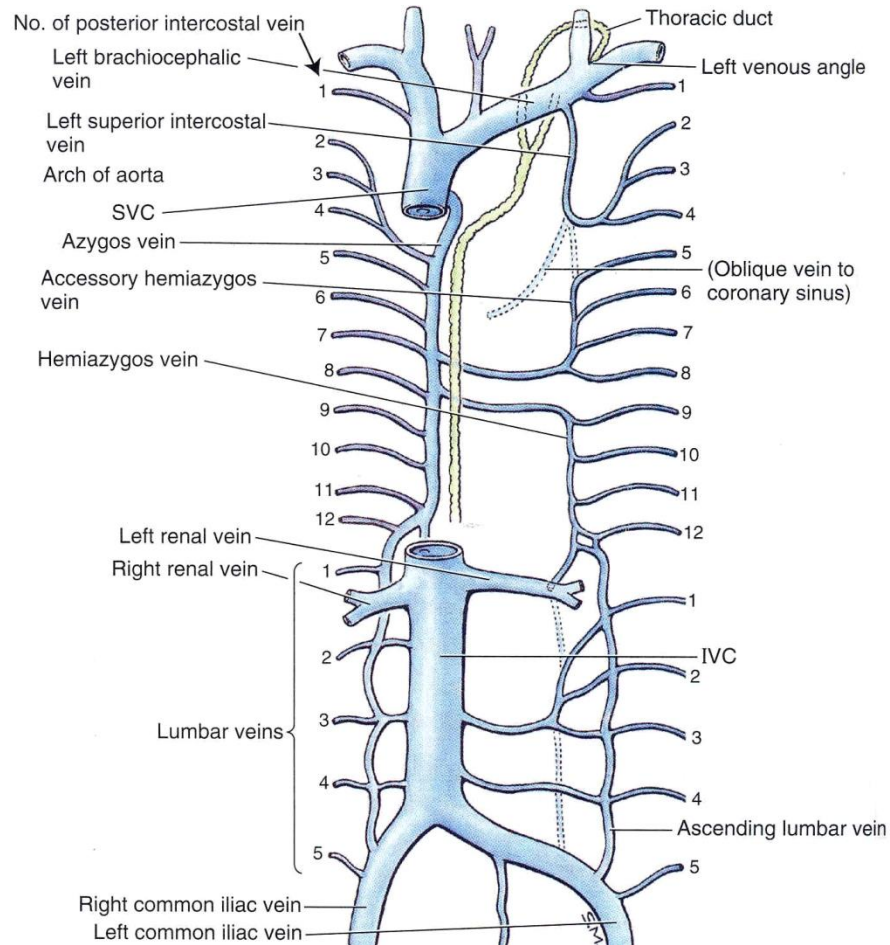
Rt.: arch of azygos vein

Lt.: brachiocephalic vein

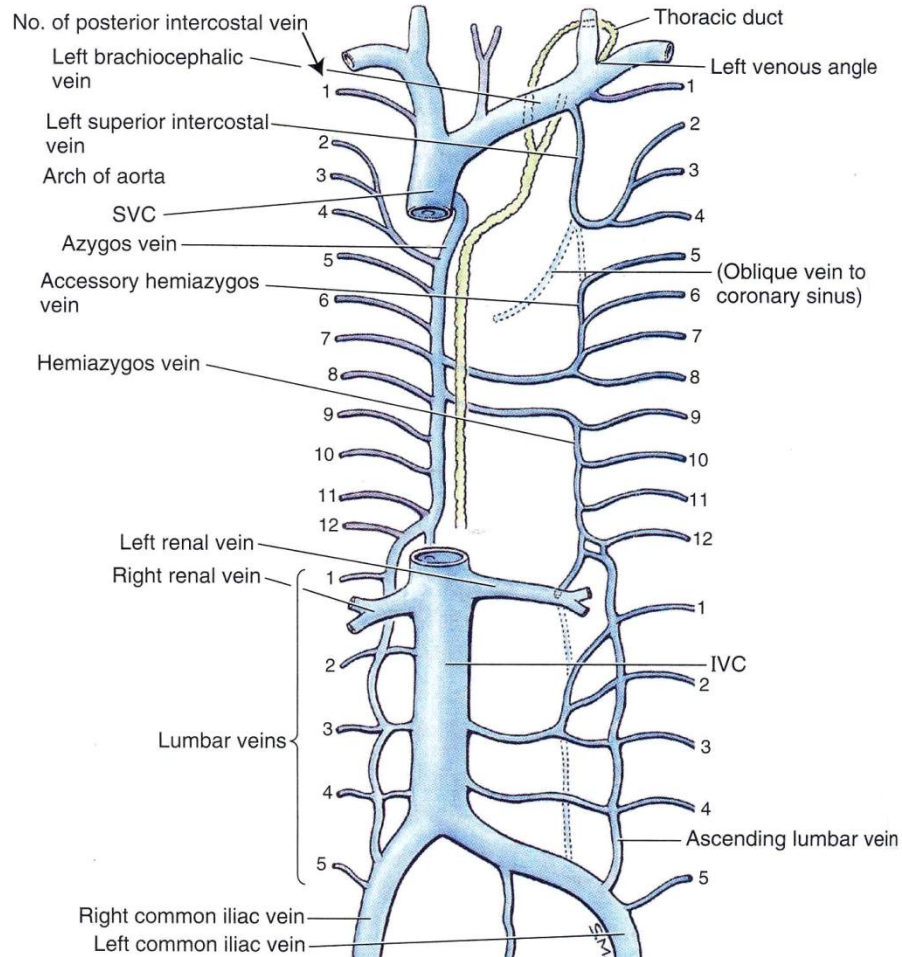
- 5th - 12th

Rt.: azygos vein

Lt.: accessory hemiazygos vein (5th - 8th)
 hemiazygos vein (9th - 12th)



Azygos vein



Superior vena cava



Ant^r to root of Rt. Lung (T4)



+ Rt. Superior intercostal v.
+ Rt. Post^r intercostal v. 5 - 12



+ Accessory hemiazygos v. (T8)
+ Hemiazygos v. (T9)



Rt. Side of vertebral column
Aortic opening of diaphragm (T12)



Rt. subcostal v. + Rt. ascending lumbar v.
+ Bl. from post^r part of IVC + renal v.

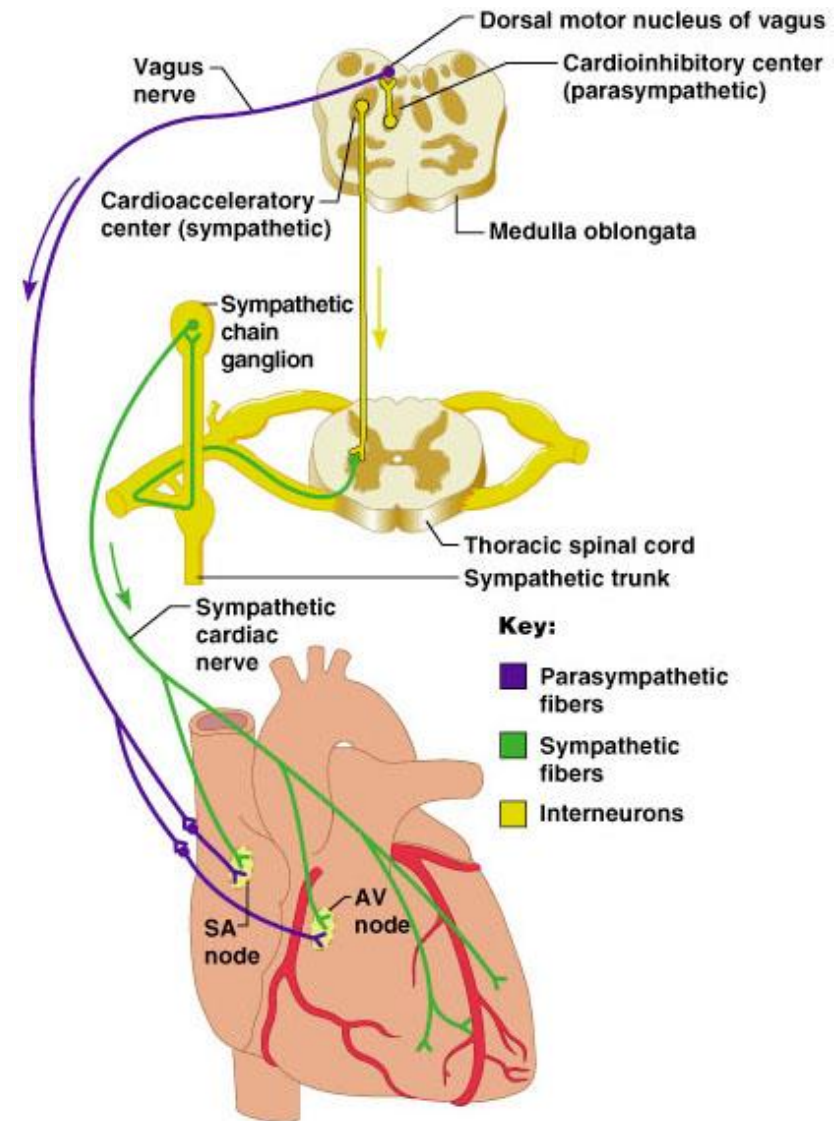
Cardiac plexuses

1. Sympathetic fiber

- Postganglionic fibers from sympathetic ganglion
- Cardioaccelerator center

2. Parasympathetic fiber

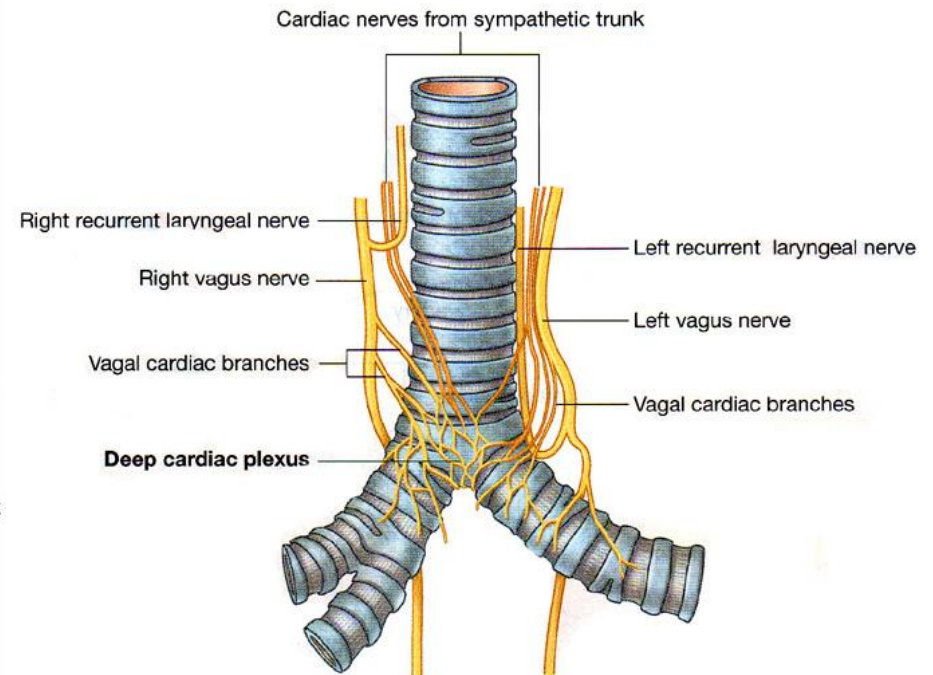
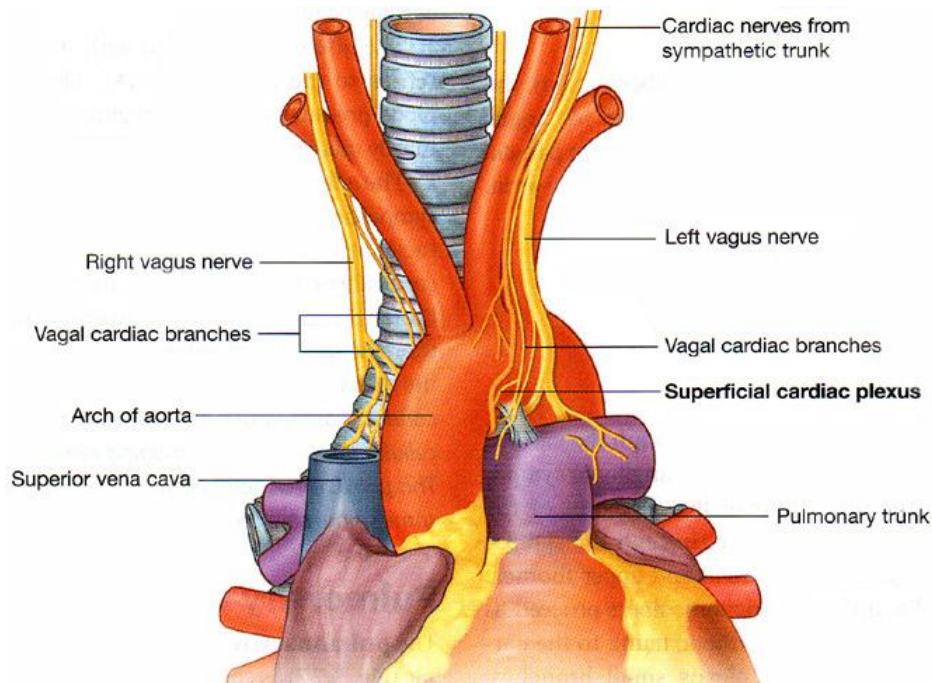
- Vagus nerve (preganglionic fiber)
- Synapse at cardiac ganglion in cardiac plexuses
- Cardioinhibitory center



Cardiac plexuses

1. Superficial cardiac plexus

2. Deep cardiac plexus



Vagus nerve

↓ Jugular foramen

Carotid sheath

(btw. int. jugular v. & common carotid a.)

Rt. vagus n.

Ant^r to Rt. subclavian a.



① **Rt. Recurrent laryngeal n.**
(hook 1st part of Rt. Subclavian a.)



Tracheoesophageal groove



Larynx

Lt. vagus n.

Ant^r to aortic arch

Lateral to
ligamentum arteriosum



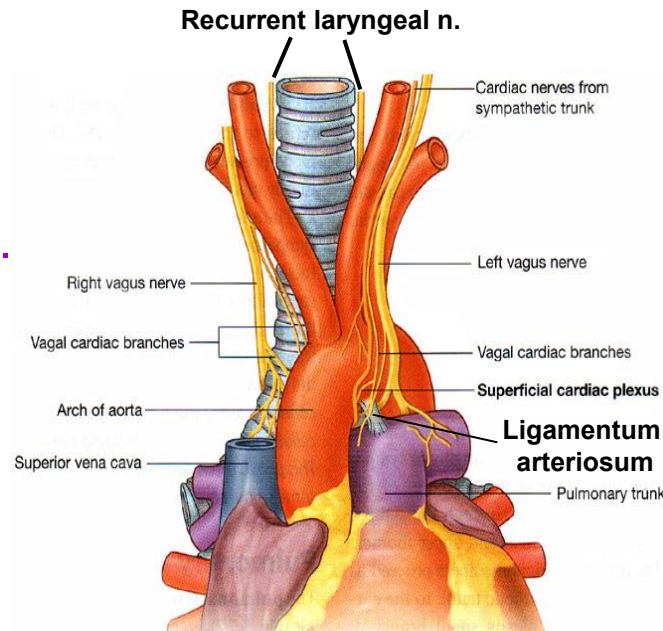
① **Lt. Recurrent laryngeal n.**
(hook aortic arch)



Tracheoesophageal groove



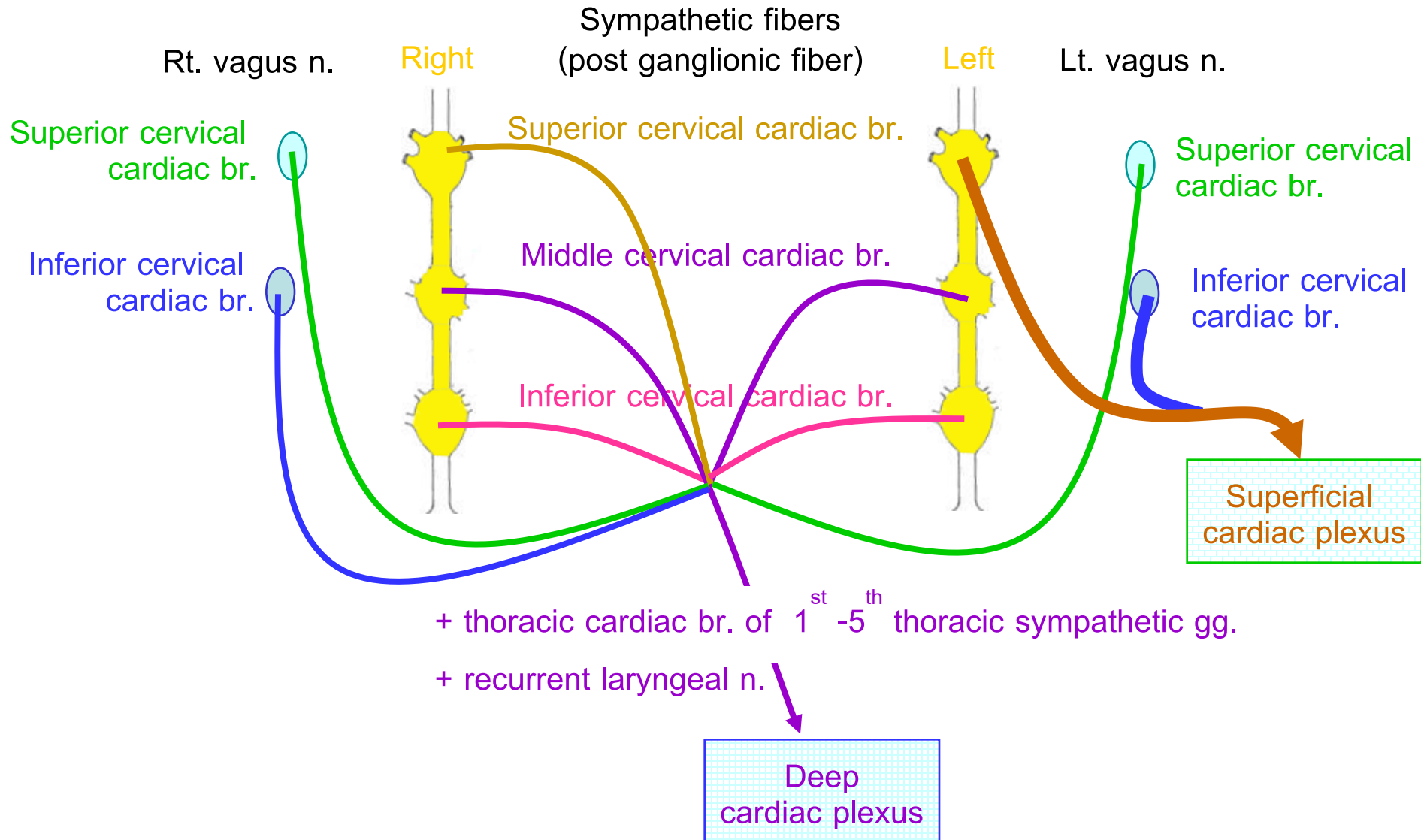
Larynx



Vagus nerve (parasympathetic fiber) + sympathetic fiber

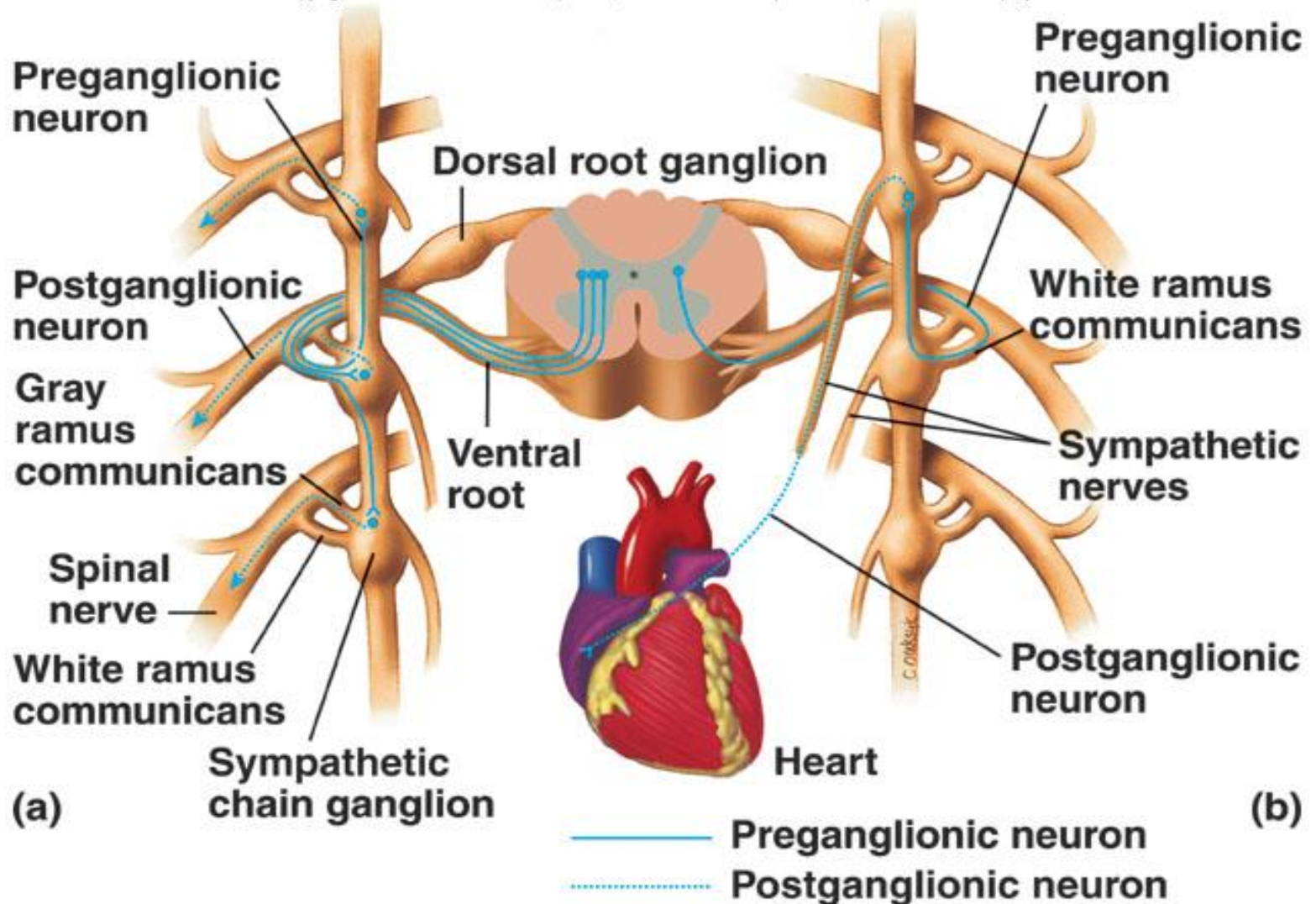
- ② Cardiac plexus --> coronary plexus --> atrium, ventricle
- ③ Pulmonary plexus --> anterior & posterior to root of lung
- ④ Esophageal plexus --> anterior & posterior esophageal plexus

Cardiac plexuses



Sympathetic nervous system

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Thoracic sympathetic trunk

- lateral to thoracic vertebrae
- 1st thoracic gg.+ inf^r cervical gg.---> stellate gg.
- continuous with
 - cervical sympathetic trunk
 - lumbar sympathetic trunk

Distributions

1. post gg. fibers ----> spinal nerves

: sweat gl., arrector pili m., vessels

2. post gg. fibers from 5 upper thoracic gg.

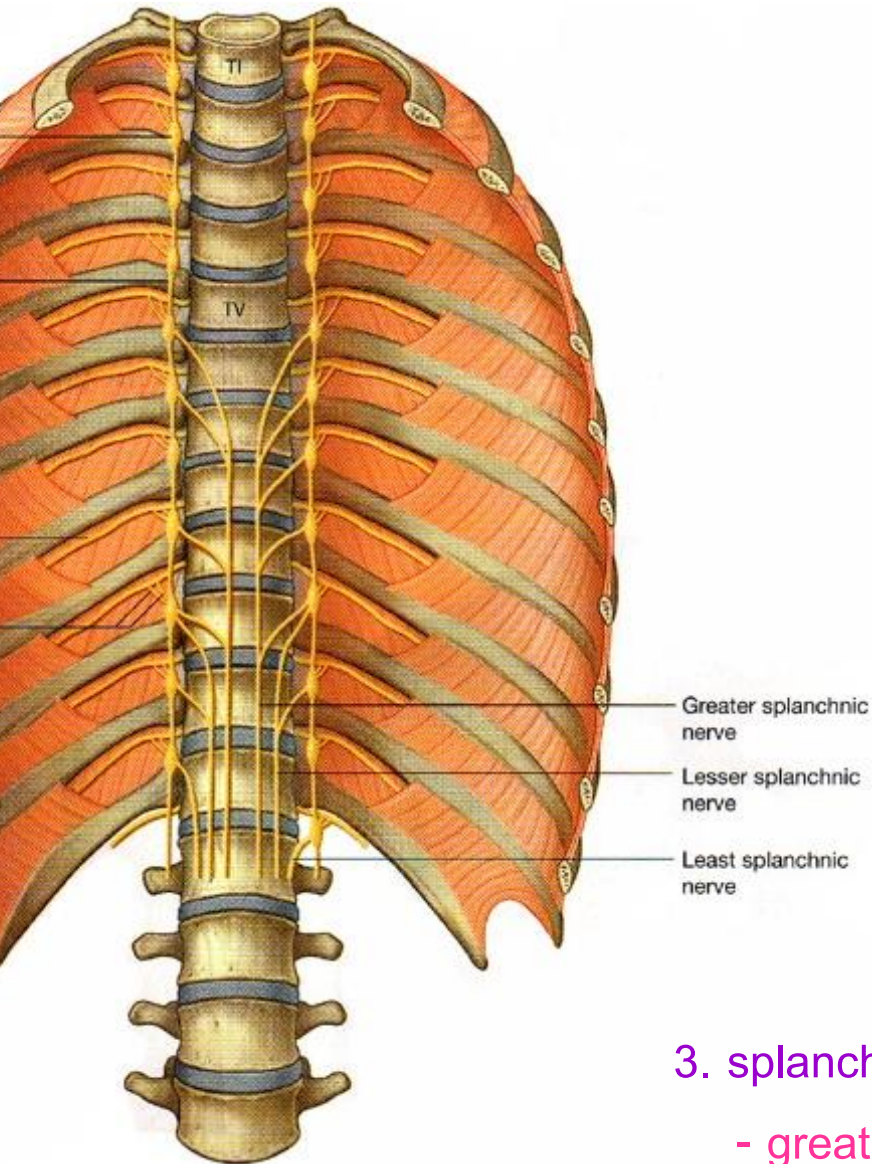
: cardiac, pulmonary, esophageal plexuses

3. splanchnic nerve

- greater splanchnic n. (th. gg. 5-9) ----> celiac gg.

- lesser splanchnic n. (th. gg. 10-11) ----> aorticorenal gg.

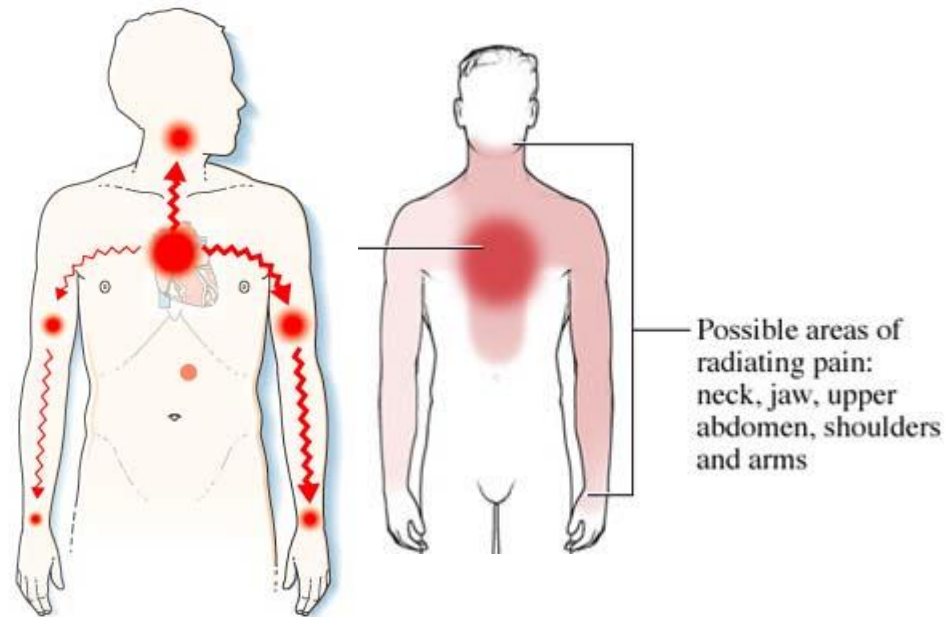
- lowest (least) splanchnic n. (th. gg.12) ---> renal plexus



Angina pectoris

The most common symptom of coronary artery disease
also known as chest discomfort

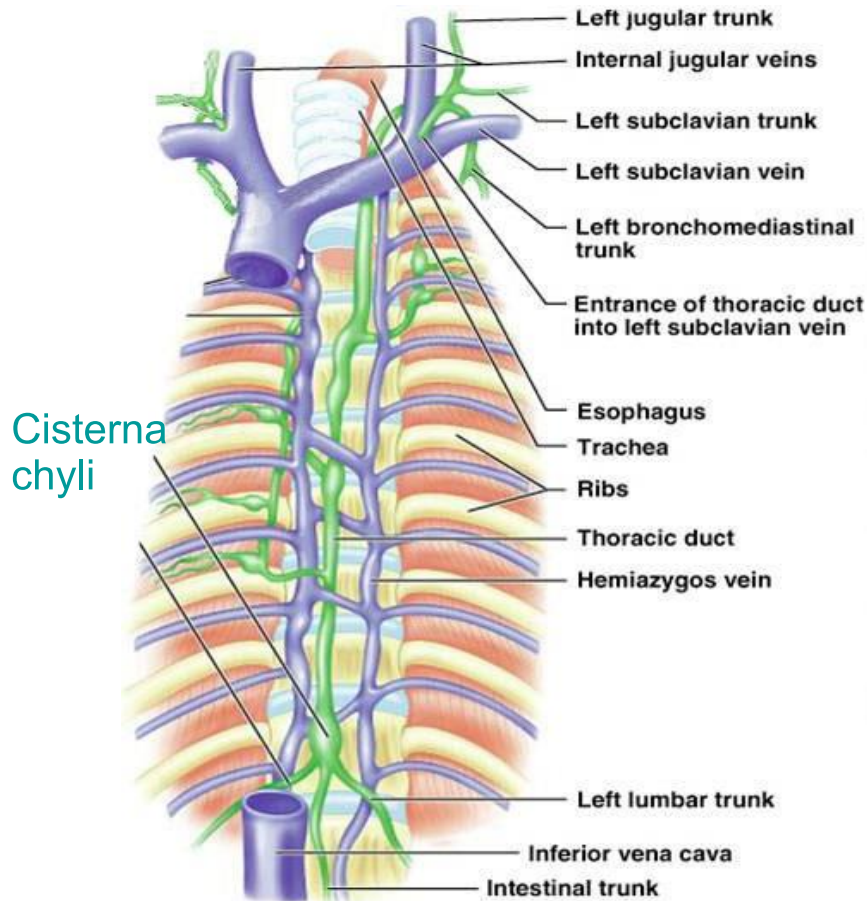
- subcostal region
- left pectoral region
- shoulder region
- medial aspect of arm
- neck, jaw



Referred pain:

- A painful body part that is not the reason for the pain
- A pain perceived in another area of the body, often far from the origin

Thoracic duct



- Cisterna chyli

- Saclike expansion located anterior to L1-2 vertebrae

- Received lymph from

- lower limbs
- pelvic and abdominal cavity
- left side of thorax, arm, head regions

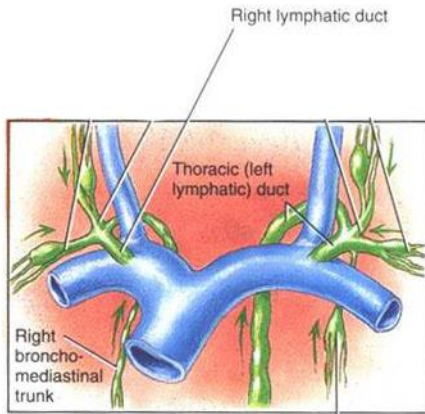
- aortic hiatus of diaphragm

- between thoracic aorta and azygos v.

- T4: cross to left side --> sup^r mediastinum

- Lt. internal jugular v. – Lt. subclavian v.

Lymphatic Circulation



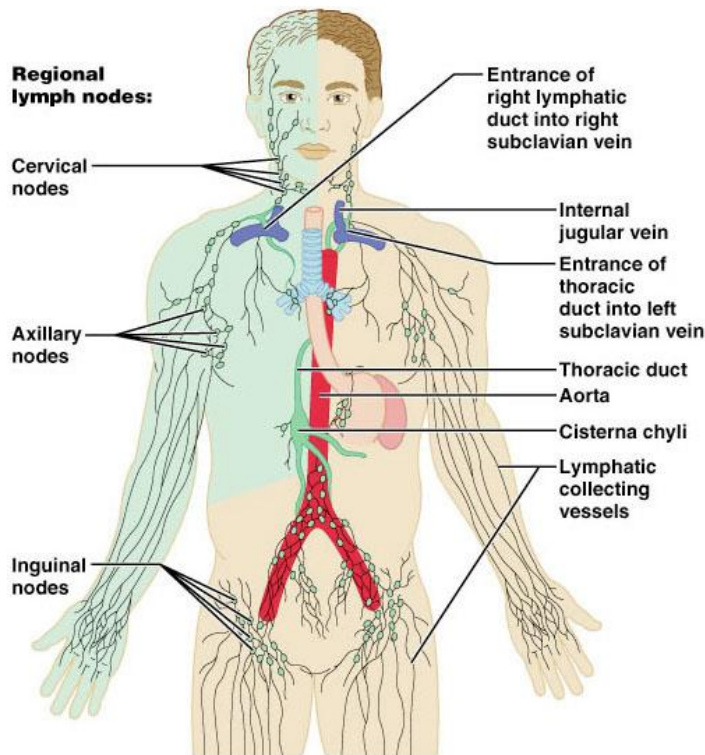
Lymph is delivered into

1. Right lymphatic duct:

- Right arm and right side of the head and thorax
- Right int. jugular v. – right subclavian v.

2. Thoracic duct:

- The rest of the body
- Cisterna chyli, left side of thorax, arm and head region
- Left int. jugular v. – left subclavian v.



THE END